

BRAF fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes BRAF gene fusions from the msk_impact_50k_2026 study, covering 179 fusion events. 151/179 fusions (84.4%) were in-frame. 163/179 fusions (91.1%) retained the Protein kinase domain. Domain retention was tested with Fisher's exact test ($p=0.0133676$, statistically significant at $\alpha=0.05$). Compared to the literature benchmark (PMC5461196: 100.0% in-frame, 100.0% domain-retained), this run observed 84.4% in-frame and 91.1% domain retention.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 179 analyzed fusion events, identifying 67 distinct fusion partner genes. The most frequent partner was KIAA1549, observed in 43 events.

Domain retention

Protein kinase domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 142/151 (94.0%) of in-frame fusions retained the domain, $p=0.0133676$ (statistically significant at $\alpha=0.05$). The Protein kinase domain appears to be required for retention. A breakpoint-position permutation test produced a corroborating empirical p-value of 0.00699301.

Domain disruption

Disruption of the RAS-binding domain and Cysteine-rich domain was tested with Fisher's exact test comparing in-frame fusions against all others; 142/151 (94.0%) of in-frame fusions disrupted the domain, $p=0.041705$ (statistically significant at $\alpha=0.05$). The RAS-binding domain and Cysteine-rich domain appear to require loss or disruption rather than retention. A breakpoint-position permutation test produced a corroborating empirical p-value of 0.963504.

Cutpoint detection

Cutpoint detection scanned 178 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 380 aa (exon 8; permutation-corrected $p=0.001998$, statistically significant at $\alpha=0.05$). This is -78 aa from the nearest configured domain boundary at 458 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" ($n=163$) and "not_retained" ($n=15$). For Frame_status, retained: 142/152 (93.4%, 95% CI 88.3%-96.4%); not_retained: 9/11 (81.8%, 95% CI 52.3%-94.9%). Welch's t-test comparing Tumor_variant_count between groups gave means of 52.0736 vs 36.4, $p=0.11965$ (not statistically significant at $\alpha=0.05$).

Composite evidence score

Composite evidence score produced results for this run (67 row(s) in composite_evidence_ranking); see the accompanying table.

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Joint partner

Joint partner reported: BRAF has no gene_pair configured; joint-partner analysis was skipped.

Mutation cooccurrence

Mutation cooccurrence produced results for this run (1 row(s) in cooccurrence_results); see the accompanying table.

Window detection

Window detection produced results for this run (57 row(s) in window_scan, 28 row(s) in top_windows); see the accompanying table.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
KIAA1549	43	0.24022346368715083	0.21553360374650618	0.001614663595055688	0.8847211802950737	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.33366411984501465	1
CUL1	5	0.027932960893854747	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.28565588873695824	2
FAM131B	3	0.01675977653631285	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.28230393342969573	3
SND1	17	0.09497206703910614	0.21553360374650618	0.001614663595055688	0.8317520556609741	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.28214333215548626	4
AKAP9	2	0.0111731843575419	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2806279557760644	5
CARMIL1	2	0.0111731843575419	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2806279557760644	6
EXOC4	2	0.0111731843575419	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2806279557760644	7
MAD1L1	2	0.0111731843575419	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2806279557760644	8
ABCC1	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2789519781224331	9

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
CAPZA2	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2789519781224331	10
CCDC6	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2789519781224331	11
CMAS	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2789519781224331	12
ERC1	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2789519781224331	13
KLHL12	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2789519781224331	14
LRBA	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2789519781224331	15
MINDY4	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2789519781224331	16
OSBPL7	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2789519781224331	17
PHTF2	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2789519781224331	18
PRIM2	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2789519781224331	19

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
PRKAR1B	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2789519781224331	20
PTPRZ1	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2789519781224331	21
RGS3	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2789519781224331	22
SNX8	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2789519781224331	23
TRA2B	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2789519781224331	24
TRIM24	8	0.0446927374301676	0.21553360374650618	0.001614663595055688	0.8951612903225806	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.27657091847204573	25
NRF1	2	0.0111731843575419	0.21553360374650618	0.001614663595055688	0.9247311827956989	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2709505364212257	26
AGAP3	6	0.0335195530726257	0.21553360374650618	0.001614663595055688	0.8637992831541219	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.26851466208951436	27
GTF2IRD1	3	0.01675977653631285	0.21553360374650618	0.001614663595055688	0.8602150537634409	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.26294909472001826	28
ATF7	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.8602150537634409	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.25959713941275575	29

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
CLEC2L	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.8602150537634409	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.25959713941275575	30
DBI	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.8602150537634409	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.25959713941275575	31
DOCK4	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.8602150537634409	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.25959713941275575	32
GIPC2	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.8602150537634409	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.25959713941275575	33
GTF2I	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.8602150537634409	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.25959713941275575	34
KCND2	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.8602150537634409	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.25959713941275575	35
NFIA	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.8602150537634409	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.25959713941275575	36
OSBPL9	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.8602150537634409	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.25959713941275575	37
PAK2	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.8602150537634409	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.25959713941275575	38
PRKAR2B	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.8602150537634409	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.25959713941275575	39

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
RRBP1	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.8602150537634409	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.25959713941275575	40
SETBP1	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.8602150537634409	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.25959713941275575	41
SLC45A3	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.8602150537634409	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.25959713941275575	42
SPOCK1	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.8602150537634409	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.25959713941275575	43
ZC3HAV1	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.8602150537634409	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.25959713941275575	44
ZNF207	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.8602150537634409	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.25959713941275575	45
CDK5RAP2	4	0.0223463687150838	0.21553360374650618	0.001614663595055688	0.803763440860215	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2561573304381657	46
CLIP1	2	0.0111731843575419	0.21553360374650618	0.001614663595055688	0.7096774193548387	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2386924719050967	47
AGK	14	0.0782122905027933	0.21553360374650618	0.001614663595055688	0.41167434715821816	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.21410374291917902	48
MKRN1	10	0.055865921787709494	0.21553360374650618	0.001614663595055688	0.4354838709677419	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.21097126087608245	49

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
TMPRSS2	4	0.0223463687150838	0.21553360374650618	0.001614663595055688	0.4408602150537635	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.20172184656719797	50
ABCC2	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.4301075268817204	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.19508101038049763	51
ITGB5	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.4301075268817204	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.19508101038049763	52
MKLN1	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.4301075268817204	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.19508101038049763	53
NEO1	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.4301075268817204	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.19508101038049763	54
PWWP2A	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.4301075268817204	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.19508101038049763	55
TBXAS1	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.4301075268817204	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.19508101038049763	56
UBE2H	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.4301075268817204	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.19508101038049763	57
WEE2	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.4301075268817204	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.19508101038049763	58
PJA2	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.3763440860215054	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.1870164942514654	59

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
SVOPL	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.3763440860215054	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.1870164942514654	60
EPB41	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.3655913978494624	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.18540359102565895	61
PLPP1	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.3655913978494624	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.18540359102565895	62
RBM33	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.3655913978494624	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.18540359102565895	63
RBMS3	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.3655913978494624	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.18540359102565895	64
SCRIB	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.3655913978494624	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.18540359102565895	65
TRIM33	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.3655913978494624	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.18540359102565895	66
LUC7L2	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.0	0.7468257003897063	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.1305648813482396	67

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
238	1	162	0	15	None	1.0	-0.0
287	1	162	1	14	0.08641975308641975	0.16187392877547135	0.7908230925727642
303	2	161	1	14	0.17391304347826086	0.23330512802756184	0.6320757153442778

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
327	19	144	6	9	0.19791666666666666	0.00883450422715351	2.053817817119616
380	19	144	9	6	0.08796296296296297	4.496509404813937e-05	4.347124493887456
381	114	49	9	6	1.5510204081632653	0.5596655980047007	0.2520713878510084
393	140	23	12	3	1.5217391304347827	0.46313499579777323	0.33429240123968906
400	141	22	12	3	1.6022727272727273	0.44662922460047316	0.3500528628421283
438	141	22	14	1	0.4577922077922078	0.6965277300948016	0.15706158880581256
439	163	0	14	1	None	0.08426966292134831	1.0743287432532127

domain_disruption tables

frame_domain_contingency_table

142	22
9	5

domain_disruption_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
ras_binding	RAS-binding domain	178	14	0	164	0.0
cysteine_rich	Cysteine-rich domain	178	14	1	163	0.005699893955461294

domain_retention tables

frame_domain_contingency_table

142	21
9	6

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
kinase	Protein kinase domain	178	163	0	15	0.0

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
11	163	179	0.9106145251396648

frequency tables

Partner_gene_counts

Partner_gene	Event_count
ABCC1	1
ABCC2	1
AGAP3	6
AGK	14
AKAP9	2
ATF7	1
CAPZA2	1
CARMIL1	2
CCDC6	1
CDK5RAP2	4
CLEC2L	1
CLIP1	2
CMAS	1
CUL1	5
DBI	1
DOCK4	1
EPB41	1
ERC1	1
EXOC4	2
FAM131B	3
GIPC2	1
GTF2I	1
GTF2IRD1	3
ITGB5	1
KCND2	1
KIAA1549	43
KLHL12	1
LRBA	1
LUC7L2	1

Partner_gene	Event_count
MAD1L1	2
MINDY4	1
MKLN1	1
MKRN1	10
NEO1	1
NFIA	1
NRF1	2
OSBPL7	1
OSBPL9	1
PAK2	1
PHTF2	1
PJA2	1
PLPP1	1
PRIM2	1
PRKAR1B	1
PRKAR2B	1
PTPRZ1	1
PWWP2A	1
RBM33	1
RBMS3	1
RGS3	1
RRBP1	1
SCRIB	1
SETBP1	1
SLC45A3	1
SND1	17
SNX8	1
SPOCK1	1
SVOPL	1
TBXAS1	1
TMPRSS2	4
TRA2B	1
TRIM24	8
TRIM33	1
UBE2H	1

Partner_gene	Event_count
WEE2	1
ZC3HAV1	1
ZNF207	1

mutation_cooccurrence tables

cooccurrence_results

comparator_gene	alteration_type	protein_change	label	contingency_table	odds_ratio	p_value	direction	cohort_sample_count	fusion_positive_sample_count	comparator_altered_sample_count	both_altered_sample_count
BRAF	point_mutation	V600E	BRAF point_mutation (V600E)	[2, 175], [1407, 52747]	0.42844552746471726	0.3375200112983454	mutually_exclusive	54331	177	1409	2

window_detection tables

window_scan

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
127	327	200	25	153	19	144	0.19791666666666666	0.00883450422715351	2.053817817119616	EVT-P-0000913-T02-IM5-23, EVT-P-0001832-T02-IM6-22, EVT-P-0004477-T01-IM5-20, EVT-P-0006412-T01-IM5-184, EVT-P-0007266-T02-IM6-120, ... (25 total)
180	380	200	28	150	19	144	0.08796296296296297	4.496509404813937e-05	4.347124493887456	EVT-P-0000913-T02-IM5-23, EVT-P-0001832-T02-IM6-22, EVT-P-0002197-T01-IM3-228, EVT-P-0004477-T01-IM5-20, EVT-P-0006412-T01-IM5-184, ... (28 total)
181	381	200	123	55	114	49	1.5510204081632653	0.5596655980047007	0.2520713878510084	EVT-P-0000913-T02-IM5-23, EVT-P-0001687-T01-IM3-220, EVT-P-0001832-T02-IM6-22, EVT-P-0001988-T01-IM3-89, EVT-P-0002071-T02-IM5-227, ... (123 total)
193	393	200	152	26	140	23	1.5217391304347827	0.46313499579777323	0.33429240123968906	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135, EVT-P-0000913-T02-IM5-23, EVT-P-0001687-T01-IM3-220, EVT-P-0001832-T02-IM6-22, ... (152 total)

start_a a	end_a a	width_a a	n_events_insid e	n_events_outsid e	n_positive_insid e	n_positive_outsid e	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
200	400	200	153	25	141	22	1.6022727272727273	0.44662922460047316	0.3500528628421283	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135 , EVT-P-0000913-T02-IM5-23, EVT-P-0001687-T01-IM3-220 , EVT-P-0001832-T02-IM6-22, ... (153 total)
227	327	100	25	153	19	144	0.19791666666666666	0.00883450422715351	2.053817817119616	EVT-P-0000913-T02-IM5-23, EVT-P-0001832-T02-IM6-22, EVT-P-0004477-T01-IM5-20, EVT-P-0006412-T01-IM5-184 , EVT-P-0007266-T02-IM6-12 0, ... (25 total)
238	338	100	25	153	19	144	0.19791666666666666	0.00883450422715351	2.053817817119616	EVT-P-0000913-T02-IM5-23, EVT-P-0001832-T02-IM6-22, EVT-P-0004477-T01-IM5-20, EVT-P-0006412-T01-IM5-184 , EVT-P-0007266-T02-IM6-12 0, ... (25 total)
238	438	200	155	23	141	22	0.4577922077922078	0.6965277300948016	0.15706158880581256	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135 , EVT-P-0000913-T02-IM5-23, EVT-P-0001687-T01-IM3-220 , EVT-P-0001832-T02-IM6-22, ... (155 total)
277	327	50	24	154	18	145	0.18620689655172415	0.007054608215979324	2.151527100175104	EVT-P-0000913-T02-IM5-23, EVT-P-0001832-T02-IM6-22, EVT-P-0004477-T01-IM5-20, EVT-P-0006412-T01-IM5-184 , EVT-P-0007266-T02-IM6-12 0, ... (24 total)
280	380	100	27	151	18	145	0.08275862068965517	3.162216908790396e-05	4.500083434297995	EVT-P-0000913-T02-IM5-23, EVT-P-0001832-T02-IM6-22, EVT-P-0002197-T01-IM3-228 , EVT-P-0004477-T01-IM5-20, EVT-P-0006412-T01-IM5-184 , ... (27 total)
281	381	100	122	56	113	50	1.5066666666666666	0.5620256581773301	0.25024385711694297	EVT-P-0000913-T02-IM5-23, EVT-P-0001687-T01-IM3-220 , EVT-P-0001832-T02-IM6-22, EVT-P-0001988-T01-IM3-89, EVT-P-0002071-T02-IM5-227 , ... (122 total)
287	337	50	24	154	18	145	0.18620689655172415	0.007054608215979324	2.151527100175104	EVT-P-0000913-T02-IM5-23, EVT-P-0001832-T02-IM6-22, EVT-P-0004477-T01-IM5-20, EVT-P-0006412-T01-IM5-184 , EVT-P-0007266-T02-IM6-12 0, ... (24 total)

start_a a	end_a a	width_a a	n_events_insid e	n_events_outsid e	n_positive_insid e	n_positive_outsid e	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
287	387	100	122	56	113	50	1.5066666666666666	0.5620256581773301	0.25024385711694297	EVT-P-0000913-T02-IM5-23, EVT-P-0001687-T01-IM3-220 , EVT-P-0001832-T02-IM6-22, EVT-P-0001988-T01-IM3-89, EVT-P-0002071-T02-IM5-227 , ... (122 total)
293	393	100	150	28	139	24	2.106060606060606	0.2608884314370491	0.5835451783717784	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135 , EVT-P-0000913-T02-IM5-23, EVT-P-0001687-T01-IM3-220 , EVT-P-0001832-T02-IM6-22, ... (150 total)
300	400	100	151	27	140	23	2.2134387351778657	0.25011484918241694	0.6018605236771493	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135 , EVT-P-0000913-T02-IM5-23, EVT-P-0001687-T01-IM3-220 , EVT-P-0001832-T02-IM6-22, ... (151 total)
302	327	25	23	155	18	145	0.2482758620689655	0.028754507085544598	1.5412940727493663	EVT-P-0000913-T02-IM5-23, EVT-P-0001832-T02-IM6-22, EVT-P-0004477-T01-IM5-20, EVT-P-0007266-T02-IM6-120 , EVT-P-0010945-T01-IM5-26, ... (23 total)
303	328	25	23	155	18	145	0.2482758620689655	0.028754507085544598	1.5412940727493663	EVT-P-0000913-T02-IM5-23, EVT-P-0001832-T02-IM6-22, EVT-P-0004477-T01-IM5-20, EVT-P-0007266-T02-IM6-120 , EVT-P-0010945-T01-IM5-26, ... (23 total)
303	353	50	23	155	18	145	0.2482758620689655	0.028754507085544598	1.5412940727493663	EVT-P-0000913-T02-IM5-23, EVT-P-0001832-T02-IM6-22, EVT-P-0004477-T01-IM5-20, EVT-P-0007266-T02-IM6-120 , EVT-P-0010945-T01-IM5-26, ... (23 total)
303	403	100	151	27	140	23	2.2134387351778657	0.25011484918241694	0.6018605236771493	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135 , EVT-P-0000913-T02-IM5-23, EVT-P-0001687-T01-IM3-220 , EVT-P-0001832-T02-IM6-22, ... (151 total)

start_a a	end_a a	width_a a	n_events_insid e	n_events_outsid e	n_positive_insid e	n_positive_outsid e	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
327	352	25	22	156	17	146	0.2328767123287671	0.023711801950937986	1.625035441061342	EVT-P-0000913-T02-IM5-23, EVT-P-0001832-T02-IM6-22, EVT-P-0004477-T01-IM5-20, EVT-P-0007266-T02-IM6-120 , EVT-P-0010945-T01-IM5-26, ... (22 total)
327	377	50	22	156	17	146	0.2328767123287671	0.023711801950937986	1.625035441061342	EVT-P-0000913-T02-IM5-23, EVT-P-0001832-T02-IM6-22, EVT-P-0004477-T01-IM5-20, EVT-P-0007266-T02-IM6-120 , EVT-P-0010945-T01-IM5-26, ... (22 total)
327	427	100	150	28	139	24	2.106060606060606	0.2608884314370491	0.5835451783717784	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135 , EVT-P-0000913-T02-IM5-23, EVT-P-0001687-T01-IM3-220 , EVT-P-0001832-T02-IM6-22, ... (150 total)
327	527	200	174	4	161	2	12.384615384615385	0.03615501809179204	1.441831416811241	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135 , EVT-P-0000913-T02-IM5-23, EVT-P-0001036-T01-IM3-192 , EVT-P-0001687-T01-IM3-22 0, ... (174 total)
331	381	50	98	80	95	68	5.588235294117647	0.005720658539439056	2.242553974077208	EVT-P-0001687-T01-IM3-220 , EVT-P-0001988-T01-IM3-89, EVT-P-0002071-T02-IM5-227 , EVT-P-0002197-T01-IM3-22 8, EVT-P-0004032-T01-IM5-1 66, ... (98 total)
338	438	100	130	48	122	41	2.6036585365853657	0.1233591284699147	0.9088287075651992	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135 , EVT-P-0001687-T01-IM3-22 0, EVT-P-0001988-T01-IM3-89, EVT-P-0002071-T02-IM5-227 , ... (130 total)
339	439	100	152	26	144	19	6.631578947368421	0.0018545198696076845	2.731768509213372	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135 , EVT-P-0001036-T01-IM3-19 2, EVT-P-0001687-T01-IM3-2 20, EVT-P-0001988-T01-IM3-89, ... (152 total)

start_a a	end_a a	width_a a	n_events_insid e	n_events_outsid e	n_positive_insid e	n_positive_outsid e	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
343	393	50	127	51	121	42	4.321428571428571	0.013112529347571712	1.8823135267889146	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135 , EVT-P-0001687-T01-IM3-22 0, EVT-P-0001988-T01-IM3-89, EVT-P-0002071-T02-IM5-227 , ... (127 total)
350	400	50	128	50	122	41	4.463414634146342	0.01259693916356655	1.8997349680458389	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135 , EVT-P-0001687-T01-IM3-22 0, EVT-P-0001988-T01-IM3-89, EVT-P-0002071-T02-IM5-227 , ... (128 total)
356	381	25	98	80	95	68	5.588235294117647	0.005720658539439056	2.242553974077208	EVT-P-0001687-T01-IM3-220 , EVT-P-0001988-T01-IM3-89, EVT-P-0002071-T02-IM5-227 , EVT-P-0002197-T01-IM3-22 8, EVT-P-0004032-T01-IM5-1 66, ... (98 total)
368	393	25	127	51	121	42	4.321428571428571	0.013112529347571712	1.8823135267889146	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135 , EVT-P-0001687-T01-IM3-22 0, EVT-P-0001988-T01-IM3-89, EVT-P-0002071-T02-IM5-227 , ... (127 total)
375	400	25	128	50	122	41	4.463414634146342	0.01259693916356655	1.8997349680458389	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135 , EVT-P-0001687-T01-IM3-22 0, EVT-P-0001988-T01-IM3-89, EVT-P-0002071-T02-IM5-227 , ... (128 total)
380	405	25	128	50	122	41	4.463414634146342	0.01259693916356655	1.8997349680458389	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135 , EVT-P-0001687-T01-IM3-22 0, EVT-P-0001988-T01-IM3-89, EVT-P-0002071-T02-IM5-227 , ... (128 total)
380	430	50	128	50	122	41	4.463414634146342	0.01259693916356655	1.8997349680458389	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135 , EVT-P-0001687-T01-IM3-22 0, EVT-P-0001988-T01-IM3-89, EVT-P-0002071-T02-IM5-227 , ... (128 total)
380	480	100	152	26	144	19	6.631578947368421	0.0018545198696076845	2.731768509213372	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135 , EVT-P-0001036-T01-IM3-19 2, EVT-P-0001687-T01-IM3-2 20, EVT-P-0001988-T01-IM3-89, ... (152 total)

start_a a	end_a a	width_a a	n_events_insid e	n_events_outsid e	n_positive_insid e	n_positive_outsid e	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
380	580	200	152	26	144	19	6.631578947368421	0.0018545198696076845	2.731768509213372	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135 , EVT-P-0001036-T01-IM3-19 2, EVT-P-0001687-T01-IM3-2 20, EVT-P-0001988-T01-IM3-89, ... (152 total)
381	406	25	125	53	122	41	11.902439024390244	3.8505600666833e-05	4.414476097457752	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135 , EVT-P-0001687-T01-IM3-22 0, EVT-P-0001988-T01-IM3-89, EVT-P-0002071-T02-IM5-227 , ... (125 total)
381	431	50	125	53	122	41	11.902439024390244	3.8505600666833e-05	4.414476097457752	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135 , EVT-P-0001687-T01-IM3-22 0, EVT-P-0001988-T01-IM3-89, EVT-P-0002071-T02-IM5-227 , ... (125 total)
381	481	100	149	29	144	19	15.157894736842104	5.1142248335696075e-0 6	5.291220183264657	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135 , EVT-P-0001036-T01-IM3-19 2, EVT-P-0001687-T01-IM3-2 20, EVT-P-0001988-T01-IM3-89, ... (149 total)
381	581	200	149	29	144	19	15.157894736842104	5.1142248335696075e-0 6	5.291220183264657	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135 , EVT-P-0001036-T01-IM3-19 2, EVT-P-0001687-T01-IM3-2 20, EVT-P-0001988-T01-IM3-89, ... (149 total)
388	438	50	32	146	27	136	0.3970588235294117 4	0.15129618600521133	0.8201720198801077	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135 , EVT-P-0003502-T01-IM5-10 1, EVT-P-0008083-T02-IM6-1 97, EVT-P-0008083-T03-IM6- 191, ... (32 total)
389	439	50	54	124	49	114	0.8596491228070176	0.7751756210061139	0.1105998941474438 4	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135 , EVT-P-0001036-T01-IM3-19 2, EVT-P-0003502-T01-IM5-1 01, EVT-P-0004103-T01-IM5- 221, ... (54 total)
393	418	25	30	148	27	136	0.7941176470588235	0.72098041318196	0.1420765335642585 3	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135 , EVT-P-0008083-T02-IM6-19 7, EVT-P-0008083-T03-IM6-1 91, EVT-P-0008567-T01-IM5- 208, ... (30 total)

start_a a	end_a a	width_a a	n_events_insid e	n_events_outsid e	n_positive_insid e	n_positive_outsid e	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
393	443	50	54	124	49	114	0.8596491228070176	0.7751756210061139	0.11059989414744384	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135, EVT-P-0001036-T01-IM3-192, EVT-P-0003502-T01-IM5-101, EVT-P-0004103-T01-IM5-221, ... (54 total)
393	493	100	54	124	49	114	0.8596491228070176	0.7751756210061139	0.11059989414744384	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135, EVT-P-0001036-T01-IM3-192, EVT-P-0003502-T01-IM5-101, EVT-P-0004103-T01-IM5-221, ... (54 total)
393	593	200	54	124	49	114	0.8596491228070176	0.7751756210061139	0.11059989414744384	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135, EVT-P-0001036-T01-IM3-192, EVT-P-0003502-T01-IM5-101, EVT-P-0004103-T01-IM5-221, ... (54 total)
400	450	50	25	153	23	140	1.0678571428571428	1.0	-0.0	EVT-P-0001036-T01-IM3-192, EVT-P-0003502-T01-IM5-101, EVT-P-0004103-T01-IM5-221, EVT-P-0005461-T02-IM5-117, EVT-P-0007392-T01-IM5-214, ... (25 total)
400	500	100	25	153	23	140	1.0678571428571428	1.0	-0.0	EVT-P-0001036-T01-IM3-192, EVT-P-0003502-T01-IM5-101, EVT-P-0004103-T01-IM5-221, EVT-P-0005461-T02-IM5-117, EVT-P-0007392-T01-IM5-214, ... (25 total)
400	600	200	25	153	23	140	1.0678571428571428	1.0	-0.0	EVT-P-0001036-T01-IM3-192, EVT-P-0003502-T01-IM5-101, EVT-P-0004103-T01-IM5-221, EVT-P-0005461-T02-IM5-117, EVT-P-0007392-T01-IM5-214, ... (25 total)
414	439	25	24	154	22	141	1.0141843971631206	1.0	-0.0	EVT-P-0001036-T01-IM3-192, EVT-P-0003502-T01-IM5-101, EVT-P-0004103-T01-IM5-221, EVT-P-0005461-T02-IM5-117, EVT-P-0007392-T01-IM5-214, ... (24 total)
438	463	25	24	154	22	141	1.0141843971631206	1.0	-0.0	EVT-P-0001036-T01-IM3-192, EVT-P-0003502-T01-IM5-101, EVT-P-0004103-T01-IM5-221, EVT-P-0005461-T02-IM5-117, EVT-P-0007392-T01-IM5-214, ... (24 total)
438	488	50	24	154	22	141	1.0141843971631206	1.0	-0.0	EVT-P-0001036-T01-IM3-192, EVT-P-0003502-T01-IM5-101, EVT-P-0004103-T01-IM5-221, EVT-P-0005461-T02-IM5-117, EVT-P-0007392-T01-IM5-214, ... (24 total)

start_a a	end_a a	width_a a	n_events_insid e	n_events_outsid e	n_positive_insid e	n_positive_outsid e	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
438	538	100	24	154	22	141	1.0141843971631206	1.0	-0.0	EVT-P-0001036-T01-IM3-192 , EVT-P-0003502-T01-IM5-10 1, EVT-P-0004103-T01-IM5-2 21, EVT-P-0005461-T02-IM5- 117, EVT-P-0007392-T01-IM 5-214, ... (24 total)
438	638	200	24	154	22	141	1.0141843971631206	1.0	-0.0	EVT-P-0001036-T01-IM3-192 , EVT-P-0003502-T01-IM5-10 1, EVT-P-0004103-T01-IM5-2 21, EVT-P-0005461-T02-IM5- 117, EVT-P-0007392-T01-IM 5-214, ... (24 total)
439	464	25	22	156	22	141	None	0.22175613072495187	0.6541243647173782	EVT-P-0001036-T01-IM3-192 , EVT-P-0004103-T01-IM5-22 1, EVT-P-0005461-T02-IM5-1 17, EVT-P-0007392-T01-IM5- 214, EVT-P-0007533-T01-IM 5-229, ... (22 total)
439	489	50	22	156	22	141	None	0.22175613072495187	0.6541243647173782	EVT-P-0001036-T01-IM3-192 , EVT-P-0004103-T01-IM5-22 1, EVT-P-0005461-T02-IM5-1 17, EVT-P-0007392-T01-IM5- 214, EVT-P-0007533-T01-IM 5-229, ... (22 total)
439	539	100	22	156	22	141	None	0.22175613072495187	0.6541243647173782	EVT-P-0001036-T01-IM3-192 , EVT-P-0004103-T01-IM5-22 1, EVT-P-0005461-T02-IM5-1 17, EVT-P-0007392-T01-IM5- 214, EVT-P-0007533-T01-IM 5-229, ... (22 total)
439	639	200	22	156	22	141	None	0.22175613072495187	0.6541243647173782	EVT-P-0001036-T01-IM3-192 , EVT-P-0004103-T01-IM5-22 1, EVT-P-0005461-T02-IM5-1 17, EVT-P-0007392-T01-IM5- 214, EVT-P-0007533-T01-IM 5-229, ... (22 total)

top_windows

start_a a	end_a a	width_a a	n_events_insid e	n_events_outsid e	n_positive_insid e	n_positive_outsid e	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
381	481	100	149	29	144	19	15.157894736842104	5.1142248335696075e-0 6	5.291220183264657	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135 , EVT-P-0001036-T01-IM3-19 2, EVT-P-0001687-T01-IM3-2 20, EVT-P-0001988-T01-IM3-89, ... (149 total)

start_a a	end_a a	width_a a	n_events_insid e	n_events_outsid e	n_positive_insid e	n_positive_outsid e	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
280	380	100	27	151	18	145	0.08275862068965517	3.162216908790396e-05	4.5000083434297995	EVT-P-0000913-T02-IM5-23, EVT-P-0001832-T02-IM6-22, EVT-P-0002197-T01-IM3-228 , EVT-P-0004477-T01-IM5-20, EVT-P-0006412-T01-IM5-184 , ... (27 total)
381	406	25	125	53	122	41	11.902439024390244	3.8505600666833e-05	4.414476097457752	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135 , EVT-P-0001687-T01-IM3-220, EVT-P-0001988-T01-IM3-89, EVT-P-0002071-T02-IM5-227 , ... (125 total)
180	380	200	28	150	19	144	0.08796296296296297	4.496509404813937e-05	4.347124493887456	EVT-P-0000913-T02-IM5-23, EVT-P-0001832-T02-IM6-22, EVT-P-0002197-T01-IM3-228 , EVT-P-0004477-T01-IM5-20, EVT-P-0006412-T01-IM5-184 , ... (28 total)
339	439	100	152	26	144	19	6.631578947368421	0.0018545198696076845	2.731768509213372	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135 , EVT-P-0001036-T01-IM3-192, EVT-P-0001687-T01-IM3-220, EVT-P-0001988-T01-IM3-89, ... (152 total)
356	381	25	98	80	95	68	5.588235294117647	0.005720658539439056	2.242553974077208	EVT-P-0001687-T01-IM3-220 , EVT-P-0001988-T01-IM3-89, EVT-P-0002071-T02-IM5-227 , EVT-P-0002197-T01-IM3-228, EVT-P-0004032-T01-IM5-166, ... (98 total)
277	327	50	24	154	18	145	0.18620689655172415	0.007054608215979324	2.151527100175104	EVT-P-0000913-T02-IM5-23, EVT-P-0001832-T02-IM6-22, EVT-P-0004477-T01-IM5-20, EVT-P-0006412-T01-IM5-184 , EVT-P-0007266-T02-IM6-120, ... (24 total)
227	327	100	25	153	19	144	0.19791666666666666	0.00883450422715351	2.053817817119616	EVT-P-0000913-T02-IM5-23, EVT-P-0001832-T02-IM6-22, EVT-P-0004477-T01-IM5-20, EVT-P-0006412-T01-IM5-184 , EVT-P-0007266-T02-IM6-120, ... (25 total)
375	400	25	128	50	122	41	4.463414634146342	0.01259693916356655	1.8997349680458389	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135 , EVT-P-0001687-T01-IM3-220, EVT-P-0001988-T01-IM3-89, EVT-P-0002071-T02-IM5-227 , ... (128 total)

start_a a	end_a a	width_a a	n_events_insid e	n_events_outsid e	n_positive_insid e	n_positive_outsid e	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
368	393	25	127	51	121	42	4.321428571428571	0.013112529347571712	1.8823135267889146	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135 , EVT-P-0001687-T01-IM3-220, EVT-P-0001988-T01-IM3-89, EVT-P-0002071-T02-IM5-227 , ... (127 total)
327	352	25	22	156	17	146	0.2328767123287671	0.023711801950937986	1.625035441061342	EVT-P-0000913-T02-IM5-23, EVT-P-0001832-T02-IM6-22, EVT-P-0004477-T01-IM5-20, EVT-P-0007266-T02-IM6-120 , EVT-P-0010945-T01-IM5-26, ... (22 total)
302	327	25	23	155	18	145	0.2482758620689655	0.028754507085544598	1.5412940727493663	EVT-P-0000913-T02-IM5-23, EVT-P-0001832-T02-IM6-22, EVT-P-0004477-T01-IM5-20, EVT-P-0007266-T02-IM6-120 , EVT-P-0010945-T01-IM5-26, ... (23 total)
327	527	200	174	4	161	2	12.384615384615385	0.03615501809179204	1.441831416811241	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135 , EVT-P-0000913-T02-IM5-23, EVT-P-0001036-T01-IM3-192 , EVT-P-0001687-T01-IM3-220, ... (174 total)
338	438	100	130	48	122	41	2.6036585365853657	0.1233591284699147	0.9088287075651992	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135 , EVT-P-0001687-T01-IM3-220, EVT-P-0001988-T01-IM3-89, EVT-P-0002071-T02-IM5-227 , ... (130 total)
388	438	50	32	146	27	136	0.39705882352941174	0.15129618600521133	0.8201720198801077	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135 , EVT-P-0003502-T01-IM5-101, EVT-P-0008083-T02-IM6-197, EVT-P-0008083-T03-IM6-191, ... (32 total)
439	464	25	22	156	22	141	None	0.22175613072495187	0.6541243647173782	EVT-P-0001036-T01-IM3-192 , EVT-P-0004103-T01-IM5-221, EVT-P-0005461-T02-IM5-117, EVT-P-0007392-T01-IM5-214, EVT-P-0007533-T01-IM5-229, ... (22 total)
300	400	100	151	27	140	23	2.2134387351778657	0.25011484918241694	0.6018605236771493	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135 , EVT-P-0000913-T02-IM5-23, EVT-P-0001687-T01-IM3-220 , EVT-P-0001832-T02-IM6-22, ... (151 total)

start_a a	end_a a	width_a a	n_events_insid e	n_events_outsid e	n_positive_insid e	n_positive_outsid e	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
293	393	100	150	28	139	24	2.106060606060606	0.2608884314370491	0.5835451783717784	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135 , EVT-P-0000913-T02-IM5-23, EVT-P-0001687-T01-IM3-220 , EVT-P-0001832-T02-IM6-22, ... (150 total)
327	427	100	150	28	139	24	2.106060606060606	0.2608884314370491	0.5835451783717784	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135 , EVT-P-0000913-T02-IM5-23, EVT-P-0001687-T01-IM3-220 , EVT-P-0001832-T02-IM6-22, ... (150 total)
200	400	200	153	25	141	22	1.6022727272727273	0.44662922460047316	0.3500528628421283	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135 , EVT-P-0000913-T02-IM5-23, EVT-P-0001687-T01-IM3-220 , EVT-P-0001832-T02-IM6-22, ... (153 total)
193	393	200	152	26	140	23	1.5217391304347827	0.4631349957977323	0.33429240123968906	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135 , EVT-P-0000913-T02-IM5-23, EVT-P-0001687-T01-IM3-220 , EVT-P-0001832-T02-IM6-22, ... (152 total)
181	381	200	123	55	114	49	1.5510204081632653	0.5596655980047007	0.2520713878510084	EVT-P-0000913-T02-IM5-23, EVT-P-0001687-T01-IM3-220 , EVT-P-0001832-T02-IM6-22, EVT-P-0001988-T01-IM3-89, EVT-P-0002071-T02-IM5-227 , ... (123 total)
281	381	100	122	56	113	50	1.5066666666666666	0.5620256581773301	0.25024385711694297	EVT-P-0000913-T02-IM5-23, EVT-P-0001687-T01-IM3-220 , EVT-P-0001832-T02-IM6-22, EVT-P-0001988-T01-IM3-89, EVT-P-0002071-T02-IM5-227 , ... (122 total)
238	438	200	155	23	141	22	0.4577922077922078	0.6965277300948016	0.15706158880581256	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135 , EVT-P-0000913-T02-IM5-23, EVT-P-0001687-T01-IM3-220 , EVT-P-0001832-T02-IM6-22, ... (155 total)

start_a a	end_a a	width_a a	n_events_insid e	n_events_outsid e	n_positive_insid e	n_positive_outsid e	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
393	418	25	30	148	27	136	0.7941176470588235	0.72098041318196	0.14207653356425853	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135, EVT-P-0008083-T02-IM6-197, EVT-P-0008083-T03-IM6-191, EVT-P-0008567-T01-IM5-208, ... (30 total)
389	439	50	54	124	49	114	0.8596491228070176	0.7751756210061139	0.11059989414744384	EVT-P-0000828-T03-IM6-94, EVT-P-0000828-T05-IM6-135, EVT-P-0001036-T01-IM3-192, EVT-P-0003502-T01-IM5-101, EVT-P-0004103-T01-IM5-221, ... (54 total)
414	439	25	24	154	22	141	1.0141843971631206	1.0	-0.0	EVT-P-0001036-T01-IM3-192, EVT-P-0003502-T01-IM5-101, EVT-P-0004103-T01-IM5-221, EVT-P-0005461-T02-IM5-117, EVT-P-0007392-T01-IM5-214, ... (24 total)
400	450	50	25	153	23	140	1.0678571428571428	1.0	-0.0	EVT-P-0001036-T01-IM3-192, EVT-P-0003502-T01-IM5-101, EVT-P-0004103-T01-IM5-221, EVT-P-0005461-T02-IM5-117, EVT-P-0007392-T01-IM5-214, ... (25 total)

Per-event results (results.tsv)

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0021458-T01-IM6-1	P-0021458-T01-IM6		ABCC1-BRAF	ABCC1	in-frame	threonine - prime	140493152	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		ABCC1 (NM_004996) - BRAF (NM_004333) rearrangement: t(7;16)(q34;p13.11)(chr7:g.140493152::chr16:g.16116105) Protein Fusion: in frame {ABCC1:BRAF}	NA
EVT-P-0034621-T01-IM6-2	P-0034621-T01-IM6		ABCC2-BRAF	ABCC2	unknown	threonine - prime	140495500	8	327	True	327	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		ABCC2 (NM_000392) - BRAF (NM_004333) rearrangement: t(7;10)(q34;24.2)(chr7:g.140495500::chr10:g.101559038) Protein Fusion: mid-exon {ABCC2:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-005258-T01-IM6-3	P-0052582-T01-IM6		AGAP3-BRAF	AGAP3	unknown	threonine - prime	140485658	10	393	True	393	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGAP3 (NM_031946) - BRAF (NM_004333) fusion: c.1637:AGAP3_c.1177+1690:BRAFInv Protein Fusion: mid-exon {AGAP3:BRAF}	NA
EVT-P-0050574-T02-IM6-4	P-0050574-T02-IM6		AGAP3-BRAF	AGAP3	in-frame	threonine - prime	140493208	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGAP3 (NM_031946) - BRAF (NM_004333) fusion: c.1222-2059:AGAP3_c.1140+900:BRAFInv Protein Fusion: in frame {AGAP3:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003979-T01-IM6-5	P-003979-T01-IM6		AGAP3-BRAF	AGAP3	in-frame	threonine - prime	140482317	11	439	True	439	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGAP3 (NM_031946) - BRAF (NM_004333) fusion: c.1221+416:AGAP3_c.1314+504:BRAFInv Protein Fusion: in frame {AGAP3:BRAF}	NA
EVT-P-005093-T01-IM6-6	P-005093-T01-IM6		AGK-BRAF	AGK	in-frame	threonine - prime	140499251	8	327	True	327	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) fusion: c.102-14294:AGK_c.980+911:BR AFInv Protein Fusion: in frame {AGK:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint exon	breakpoint protein position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0042835-T03-IM6-7	P-0042835-T03-IM6		AGK-BRAF	AGK	in-frame	threonine - prime	140497995	8	327	True	327	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) fusion: c.102-13644:AGK_c.980+2167:BRAFInv Protein Fusion: in frame {AGK:BRAF}	NA
EVT-P-0042835-T02-IM6-8	P-0042835-T02-IM6		AGK-BRAF	AGK	in-frame	threonine - prime	140497995	8	327	True	327	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) fusion: c.102-13644:AGK_c.980+2167:BRAFInv Protein Fusion: in frame {AGK:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint exon	breakpoint protein position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0050635-T01-IM6-9	P-0050635-T01-IM6		AKAP9-BRAF	AKAP9	in-frame	threonine - prime	140487501	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AKAP9 (NM_005751) - BRAF (NM_004333) fusion: c.6613-1489: AKAP9_c.1141-117: BRAFInv Protein Fusion: in frame {AKAP9:BRAF}	NA
EVT-P-0030237-T03-IM6-10	P-0030237-T03-IM6		AKAP9-BRAF	AKAP9	in-frame	threonine - prime	140492357	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AKAP9 (NM_005751) - BRAF (NM_004333) fusion: c.3318+1378: AKAP9_c.1140+1751: BRAFInv Protein Fusion: in frame {AKAP9:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint exon	breakpoint protein position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0023821-T01-IM6-12	P-0023821-T01-IM6		ATF7-BRAF	ATF7	in-frame	threonine - prime	140483233	10	393	True	393	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		ATF7 (NM_006856) - BRAF (NM_004333) Rearrangement : t(7;12)(q34;q13.13)(chr7:g.140483233::chr12:g.53913299) Protein Fusion: in frame {ATF7:BRAF}	NA
EVT-P-0007861-T01-IM5-13	P-0007861-T01-IM5		BRAF-AGAP3	AGAP3	in-frame	threonine - prime	140490697	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGAP3 (NM_031946) - BRAF (NM_004333) rearrangement: c.1326+220:AGAP3_c.141-3313:BRAFinv Protein fusion: in frame (AGAP3-BRAF)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint exon	breakpoint protein position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0021519-T02-IM6-14	P-0021519-T02-IM6		BR-AF-AG-AP3	AGAP3	in-frame	threonine - prime	140493112	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGAP3 (NM_031946) - BRAF (NM_004333) Rearrangement : c.1326+142:AGAP3_c.1140+996:BRAFInv Protein Fusion: in frame {AGAP3:BRAF}	NA
EVT-P-0009757-T01-IM5-15	P-0009757-T01-IM5		BR-AF-AG-AP3	AGAP3	in-frame	threonine - prime	140493871	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGAP3 (NM_031946) - BRAF (NM_004333) rearrangement: c.1222-2154:AGAP3_c.140+237:BRAFInv Protein fusion: in frame (AGAP3-BRAF)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint exon	breakpoint protein position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0036886-T01-IM6-16	P-0036886-T01-IM6		BRAF-AKK	AGK	in-frame	threonine - prime	140494309	8	327	True	327	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) rearrangement: c.101+10859:AGK_c.981-42:BRAFInv Protein Fusion: in frame {AGK:BRAF}	NA
EVT-P-0025561-T03-IM6-17	P-0025561-T03-IM6		BRAF-AKK	AGK	in-frame	threonine - prime	140498719	8	327	True	327	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) fusion (AGK exons 1-2 fused to BRAF exons 8-18): c.980+1288:BRAF_c.101+772:AGKInv Protein Fusion: in frame {AGK:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frameshift_status	target_role	breakpoint genomic	breakpoint exon	breakpoint protein position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0025561-T01-IM6-18	P-0025561-T01-IM6		BRAF-AGK	AGK	in-frame	five_prime	140498874	7	327	True	326	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than thisproject's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) - AGK (NM_018238) Rearrangement: c.980+1288:BRAF_c.101+772:AGKinv Protein Fusion: in frame {BRAF:AGK}	NA
EVT-P-0024227-T01-IM6-19	P-0024227-T01-IM6		BRAF-AGK	AGK	in-frame	three_prime	140495070	8	327	True	327	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) rearrangement: c.101+16796:AGK_c.981-803:BRAFinv Protein Fusion: in frame {AGK:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frameshift_status	target_role	breakpoint genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0004477-T01-IM5-20	P-0004477-T01-IM5		BRAF-AKK	AGK	in-frame	threonine - prime	140495405	8	327	True	327	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) fusion (AGK exons 1 to 2 and BRAF exons 8 to 18): c.101+10076:AGK_c.981-1138:BRAFinv Protein fusion: in frame (AGK-BRAF)	NA
EVT-P-0023378-T01-IM6-21	P-0023378-T01-IM6		BRAF-AKK	AGK	in-frame	five - prime	140498362	7	327	True	326	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than thisproject's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) - AGK (NM_018238) rearrangement: c.980+1800:BRAF_c.101+14574:AGKinv Protein Fusion: in frame {BRAF:AGK}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint_exon	breakpoint_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-000183 2-T02-IM6-22	P-0001832-T02-IM6		BR AF- AG K	AG K	in- fr a m e	t h r e e _ p r i m e	140494489	8	327	True	327	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) fusion (AGK exon 1 fused in-frame to BRAF exons 8-18): c.102-572:AGK_c.981-222:BRAFInv Protein fusion: in frame (AGK-BRAF)	NA
EVT-P-000091 3-T02-IM5-23	P-0000913-T02-IM5		BR AF- AG K	AG K	in- fr a m e	t h r e e _ p r i m e	140496386	8	327	True	327	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) Fusion (AGK exon 2 fused with BRAF exon 8) : c.101+13932:AGK_c.981-2119:BRAFInv Protein fusion: in frame (AGK-BRAF)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frameshift_status	target_role	breakpoint genomic	breakpoint_exon	breakpoint_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0029531-T01-IM6-24	P-0029531-T01-IM6		BR-AF-AK	AGK	unknown	threonine - prime	140500233	7	303	False	303	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) fusion (AGK exons 1-2 fused to BRAF exons 7-18): c.101+4572:AGK_c.909:BRAF Protein Fusion: mid-exon {AGK:BRAF}	NA
EVT-P-0030746-T01-IM6-25	P-0030746-T01-IM6		BR-AF-AK	AGK	in-frame	threonine - prime	140497533	8	327	True	327	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) fusion: c.101+8348:AGK_c.980+2629:BR Protein Fusion: in frame {AGK:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frameshift_status	target_role	breakpoint genomic	breakpoint exon	breakpoint protein position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-001094 5-T01-I M5-26	P-00109 45-T 01-I M5		BRAF-AGK	AGK	in-frame	threonine - prime	140497610	8	327	True	327	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) rearrangement : c.101+12646:AGK_c.980+2552:BRAFinv Protein fusion: in frame (AGK-BRAF)	NA
EVT-P-004903 5-T01-I M6-85	P-00490 35-T 01-I M6		BRAF-CLEC2L	CLEC2L	in-frame	five - prime	140485799	9	393	True	392	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than thisproject's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) - CLEC2L (NM_001080511) rearrangement: c.1177+1549:BRAF_c.191-5932:CLEC2Linv Protein Fusion: in frame {BRAF:CLEC2L}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint_exon	breakpoint_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-000479 5-T01-IM5-87	P-0004795-T01-IM5		BRAF-CUL1	CUL1	in-frame	threonine - prime	140493811	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CUL1(NM_003592) - BRAF (NM_004333) Rearrangement :c.790-427:CUL1_c.1140+297:BRAFInv Protein fusion: in frame (CUL1-BRAF)	NA
EVT-P-002225 7-T01-IM6-88	P-0022257-T01-IM6		BRAF-EPB41	EPB41	in-frame	threonine - prime	140482687	11	439	True	439	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		EPB41 (NM_001166005) - BRAF (NM_004333) fusion (EPB41 exons 1-5 fused in-frame with BRAF exons 11-18): t(1;7)(p35.4;q34)(chr1:g.29424227::chr7:g.140482687) Protein Fusion: in frame {EPB41:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-000198 8-T01-I M3-89	P-000198 88-T01-I M3		BR AF- FA M1 31B	F A M 13 1 B	in- fr a m e	t h r e e _ p r i m e	1404 8962 9	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	R A S- bind ing do m a i n; C ys tei ne -ri ch do m a i n		FAM131B (NM_001031690) - BRAF (NM_004333) Deletion : c.175-103: FAM131B_c.1141-2 245:BRAFdel Protein fusion: in frame (FAM131B-BRAF)	NA
EVT-P-001049 4-T01-I M5-90	P-001049 94-T01-I M5		BR AF- GIP C2	G I P C 2	in- fr a m e	t h r e e _ p r i m e	1404 8727 6	10	393	True	393	True	exact match	retained	1.0	False	Protein kinase domain	R A S- bind ing do m a i n; C ys tei ne -ri ch do m a i n		GIPC2 (NM_017655) -BRAF (NM_004333) Rearrangement : t(1; 7)(p31.1;q34)(chr1:g .78571528::chr7:g.1 40487276) Protein fusion: in frame (GIPC2-BRAF)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frameshifts	target_role	breakpoint genomic	breakpoint exon	breakpoint protein position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0054046-T01-IM6-93	P-0054046-T01-IM6		BRAF-GTF2IRD1	GTF2IRD1	in-frame	five_prime	140484706	9	393	True	392	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than thisproject's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) - GTF2IRD1 (NM_005685) rearrangement: c.1178-1749:BRAF_c.2320+10446:GTF2IRD1inv Protein Fusion: in frame {BRAF:GTF2IRD1}	NA
EVT-P-0000828-T03-IM6-94	P-0000828-T03-IM6		BRAF-GTF2IRD1	GTF2IRD1	in-frame	five_prime	140483554	9	393	True	392	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than thisproject's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) - GTF2IRD1 (NM_005685) rearrangement: c.1178-597:BRAF_c.2320+14181:GTF2IRD1inv Protein Fusion: in frame {BRAF:GTF2IRD1}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frameshift_status	target_role	breakpoint genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0015847-T02-IM6-95	P-0015847-T02-IM6		BRAF-ITGB5	ITGB5	out-of-frame	five_prime	140497019	7	327	True	326	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than thisproject's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) - ITGB5 (NM_002213) Rearrangement : t(3,7)(q21.2,q34)(chr3:g.124579522::chr7:g.140497019) Protein Fusion: out of frame {BRAF:ITGB5}	NA
EVT-P-0014377-T01-IM6-99	P-0014377-T01-IM6		BRAF-NFIA	NFIA	in-frame	three_prime	140486673	10	393	True	393	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		NFIA (NM_001145512) - BRAF (NM_004333) rearrangement: t(1;7)(p31.3;q34)(chr1:g.61836194::chr7:g.140486673) Protein fusion: in frame {NFIA-BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint_exon	breakpoint_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0010398-T01-IM5-100	P-0010398-T01-IM5		BRAF-OSBPL9	OSBPL9	in-frame	threonine - prime	140484912	10	393	True	393	True	exact match	retained	1.0	False	Protein kinase domain	RA S-binding domain; Cysteine-rich domain		OSBPL9 (NM_148909) - BRAF (NM_004333) Rearrangement : t(1;7)(p32.3;q34)(chr1:g.52228623::chr7:g.140484912) Protein fusion: in frame (OSBPL9-BRAF)	NA
EVT-P-0003502-T01-IM5-101	P-0003502-T01-IM5		BRAF-PJA2	PJA2	in-frame	five - prime	140482379	10	438	True	438	True	exact match	lost	0.0	False	RA S-binding domain; Cysteine-rich domain	Protein kinase domain		NA Protein fusion: in frame (BRAF-PJA2)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint exon	breakpoint protein position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0036870-T01-IM6-102	P-0036870-T01-IM6		BRAF-PRIM2	PRIM2	in-frame	threonine - prime	140493232	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		PRIM2 (NM_000947) - BRAF (NM_004333) Rearrangement : t(6;7)(p11.1;q34)(chr6:g.57410815::chr7:g.140493232) Protein Fusion: in frame {PRIM2:BRAF}	NA
EVT-P-0009246-T01-IM5-103	P-0009246-T01-IM5		BRAF-RBM33	RBM33	in-frame	threonine - prime	140482661	11	439	True	439	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		RBM33 (NM_053043) - BRAF (NM_004333) rearrangement: c.171+526:RBM33_c.1314+160:BRAFInv Protein fusion: in frame (RBM33-BRAF)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frameshift_status	target_role	breakpoint genomic	breakpoint_exon	breakpoint_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0016313-T01-IM6-104	P-0016313-T01-IM6		BRAF-SLC45A3	SLC45A3	unknown	threonine - prime	140486608	10	393	True	393	True	exact match	retained	1.0	False	Protein kinase domain	RA-S-binding domain; Cysteine-rich domain		SLC45A3 (NM_033102) - BRAF (NM_004333) fusion (SLC45A3 exons 1 fused to BRAF exons 10-18): t(1;7)(q32.1;q34)(chr1:g.205638452::chr7:g.140486608) Transcript Fusion {SLC45A3:BRAF}	NA
EVT-P-0064764-T01-IM7-105	P-0064764-T01-IM7		BRAF-SND1	SND1	in-frame	five - prime	140493289	8	380	True	380	True	exact match	lost	0.0	False	RA-S-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) - SND1 (NM_014390) fusion: c.1140+819:BRAF_c.1039-3950:SND1inv Protein Fusion: in frame {BRAF:SND1}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frameshift_status	target_role	breakpoint genomic	breakpoint exon	breakpoint protein position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-002998 1-T01-I M6-106	P-00299 81-T 01-I M6		BR AF- SP OC K1	S P O C K 1	in-frame	threonine - prime	140483862	10	393	True	393	True	exact match	retained	1.0	False	Protein kinase domain	RA S-binding domain; Cysteine-rich domain		SPOCK1 (NM_004598) - BRAF (NM_004333) fusion: t(5;7)(q31.2;q34)(chr5:g.136401840::chr7:g.140483862) Protein Fusion: in frame {SPOCK1:BRAF}	NA
EVT-P-006422 3-T01-I M7-107	P-00642 23-T 01-I M7		BR AF- TB XA S1	T B X A S 1	out-of-frame	five - prime	140494512	7	327	True	326	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than thisproject's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False	RA S-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) - TBXAS1 (NM_001061) rearrangement: c.981-245:BRAF_c.336+2985:TBXAS1inv Protein Fusion: out of frame {BRAF:TBXAS1}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0021469-T01-IM6-108	P-0021469-T01-IM6		BRAF-TRIM33	TRIM33	in-frame	threonine - prime	140481644	11	439	True	439	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TRIM33 (NM_015906) - BRAF(NM_004333) rearrangement: t(1;7)(p13.2;q34)(chr1:g.114963985::chr7:g.140481644) Protein Fusion: in frame {TRIM33:BRAF}	NA
EVT-P-0042835-T03-IM6-109	P-0042835-T03-IM6		BRAF-WEE2	WEE2	unknown	five - prime	140498912	7	327	True	326	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than thisproject's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) - WEE2 (NM_001105558) rearrangement: c.980+1250:BRAF_c.1305:WEE2inv Protein Fusion: mid-exon {BRAF:WEE2}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint exon	breakpoint protein position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0015985-T03-IM6-111	P-0015985-T03-IM6		CAPZA2-BRAF	CAPZA2	in-frame	threonine - prime	140491425	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CAPZA2 (NM_006136) - BRAF (NM_004333) fusion: c.40-938:CAPZA2_c.1140+2683: BRAFInv Protein Fusion: in frame {CAPZA2:BRAF}	NA
EVT-P-0033764-T04-IM6-112	P-0033764-T04-IM6		CARML1-BRAF	CARML1	in-frame	threonine - prime	140491509	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		LRRC16A (NM_017640) - BRAF (NM_004333) fusion: t(6;7)(p22.2;q34)(chr6:g.25554753::chr7:g.140491509) Protein Fusion: in frame {LRRC16A:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003079 6-T01-IM6-113	P-0030796-T01-IM6		CARM1L1-BRAF	CARM1L1	in-frame	threonine - prime	140491645	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		LRRC16A (NM_017640) - BRAF (NM_004333) fusion (LRRC16A exons 1-29 fused to BRAF exons 9-18): t(6;7)(p22.2;q34)(chr6:g.25580104::chr7:g.140491645) Protein Fusion: in frame {LRRC16A:BRAF}	NA
EVT-P-000583 4-T01-IM5-114	P-0005834-T01-IM5		CCDC6-BRAF	CCDC6	in-frame	threonine - prime	140488401	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CCDC6 (NM_005436) -BRAF (NM_004333) Rearrangement : t(10;7)(chr10q21.2;q34)(chr10:g.61627617::chr7:g.140488401) Protein fusion: in frame (CCDC6-BRAF)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint exon	breakpoint protein position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0038194-T01-IM6-115	P-0038194-T01-IM6		CDK5RAP2-BRAF	CDK5RAP2	in-frame	threonine - prime	140484954	10	393	True	393	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CDK5RAP2 (NM_018249) - BRAF (NM_004333) rearrangement: t(7;9)(q34;q33.2)(chr7:g.140484954::chr9:g.123187156) Protein Fusion: in frame {CDK5RAP2:BRAF}	NA
EVT-P-0004203-T01-IM5-116	P-0004203-T01-IM5		CDK5RAP2-BRAF	CDK5RAP2	in-frame	threonine - prime	140491585	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CDK5RAP2 (NM_018249) - BRAF (NM_004333) rearrangement: t(7;9)(7q34;q33.2)(chr7:140491585::chr9:123251876) Protein fusion: in frame (CDK5RAP2-BRAF)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frameshift_status	target_role	breakpoint genomic	breakpoint exon	breakpoint protein position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0005461-T02-IM5-117	P-0005461-T02-IM5		CDK5RAP2-BRAF	CDK5RAP2	in-frame	threonine_prime	140481684	11	439	True	439	True	exact match	retained	1.0	False	Protein kinase domain	RA-S-binding domain; Cysteine-rich domain		CDK5RAP2 (NM_018249) - BRAF (NM_004333) rearrangement : t(7;9)(7q34;q33.2)(chr7:140481684::chr9:123255452) Protein fusion: in frame (CDK5RAP2-BRAF)	NA
EVT-P-0014969-T01-IM6-118	P-0014969-T01-IM6		CDK5RAP2-BRAF	CDK5RAP2	in-frame	five_prime	140493700	8	380	True	380	True	exact match	lost	0.0	False	RA-S-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) - CDK5RAP2 (NM_018249) Rearrangement : t(7;9)(q34;q33.3)(chr7:g.140493700::chr9:g.123269531) Protein fusion: in frame (BRAF-CDK5RAP2)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frameshift_status	target_role	breakpoint genomic	breakpoint exon	breakpoint protein position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0040461-T01-IM6-119	P-0040461-T01-IM6		CLIP1-BRAF	CLIP1	in-frame	threonine - prime	140488218	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CLIP1 (NM_001247997) - BRAF (NM_004333) fusion: t(7;12)(q34;q24.31)(chr7:g.140488218::chr12:g.122833264) Protein Fusion: in frame {CLIP1:BRAF}	NA
EVT-P-0007266-T02-IM6-120	P-0007266-T02-IM6		CLIP1-BRAF	CLIP1	in-frame	threonine - prime	140495673	8	327	True	327	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CLIP1 (NM_001247997) - BRAF (NM_004333) fusion: t(7;12)(q34;q24.31)(chr7:g.140495673::chr12:g.122767026) Protein Fusion: in frame {CLIP1:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint_exon	breakpoint_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-002499 5-T01-I M6-121	P-00249 95-T 01-I M6		CMAS-BRAF	CMAS	unknown	threonine - prime	140488046	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		BRAF (NM_004333) rearrangement: t(7;12)(q34;p12.1)(chr7:g.140488046::chr12:g.22218755) Transcript Fusion {CMAS:BRAF}	NA
EVT-P-003964 8-T01-I M6-122	P-00396 48-T 01-I M6		CUL1-BRAF	CUL1	in-frame	threonine - prime	140492893	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CUL1 (NM_003592) - BRAF (NM_004333) fusion: c.789+2950:CUL1_c.1140+1215:BRAF in frame Protein Fusion: in frame {CUL1:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint exon	breakpoint protein position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003964 8-T03-IM6-123	P-0039648-T03-IM6		CUL1-BRAF	CUL1	in-frame	threonine - prime	140492893	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CUL1 (NM_003592) - BRAF (NM_004333)fusion: c.789+2950:CUL1_c.1140+1215:BRAFinv Protein Fusion: in frame {CUL1:BRAF}	NA
EVT-P-006408 8-T01-IM7-124	P-0064088-T01-IM7		CUL1-BRAF	CUL1	in-frame	threonine - prime	140489451	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CUL1 (NM_003592) - BRAF (NM_004333)fusion: c.789+1956:CUL1_c.1141-2067:BRAFinv Protein Fusion: in frame {CUL1:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0068173-T01-IM7-125	P-0068173-T01-IM7		CUL1-BRAF	CUL1	in-frame	threonine - prime	140491990	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CUL1 (NM_003592) - BRAF (NM_004333)fusion: c.789+191:CUL1_c.1140+2118:BRAFInv Protein Fusion: in frame {CUL1:BRAF}	NA
EVT-P-0050196-T01-IM6-126	P-0050196-T01-IM6		DBI-BRAF	DBI	in-frame	threonine - prime	140486284	10	393	True	393	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		DBI (NM_001178017) - BRAF (NM_004333) fusion: t(2;7)(q14.2;q34)(chr2:g.120128894::chr7:g.140486284) Protein Fusion: in frame {DBI:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint exon	breakpoint protein position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-006458-T01-IM7-127	P-0064588-T01-IM7		DOCK4-BRAF	DOCK4	out-of-frame	threonine - prime	140483872	10	393	True	393	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		DOCK4 (NM_014705) - BRAF (NM_004333) fusion: c.4650+298: DOCK4_c.1178-915: BRAFdup Protein Fusion: out of frame {DOCK4:BRAF}	NA
EVT-P-001483-T02-IM6-128	P-0014833-T02-IM6		ERC1-BRAF	ERC1	in-frame	threonine - prime	140490668	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		ERC1 (NM_178040) - BRAF (NM_004333) fusion: t(7;12)(q34;p13.33)(chr7:g.140490668::chr12:g.1246645) Protein Fusion: in frame {ERC1:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003693 7-T01-I M6-129	P-00369 37-T01-I M6		EXOC4-BRAF	EXOCC4	out-of-frame	threonine - prime	140490241	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		EXOC4 (NM_021807) - BRAF (NM_004333) fusion: c.1514+93469:EXOC4_c.1141-2857:BRAFinv Protein Fusion: out of frame {EXOC4:BRAF}	NA
EVT-P-002420 6-T01-I M6-130	P-00242 06-T01-I M6		EXOC4-BRAF	EXOCC4	out-of-frame	threonine - prime	140487956	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		EXOC4 (NM_021807) - BRAF (NM_004333) rearrangement: c.87-8072:EXOC4_c.1141-572:BRAFinv Protein Fusion: out of frame {EXOC4:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frameshift_status	target_role	breakpoint genomic	breakpoint exon	breakpoint protein position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0047918-T04-IM7-131	P-0047918-T04-IM7		FAM131B-BRAF	FAM131B	in-frame	threonine - prime	140487437	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		FAM131B (NM_001031690) - BRAF (NM_004333) fusion: c.175-20:FAM131B_c.1141-53:BRAFdel Protein Fusion: in frame {FAM131B:BRAF}	NA
EVT-P-0047918-T02-IM6-132	P-0047918-T02-IM6		FAM131B-BRAF	FAM131B	in-frame	threonine - prime	140487437	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		FAM131B (NM_001031690) - BRAF (NM_004333) fusion: c.175-20:FAM131B_c.1141-53:BRAFdel Protein Fusion: in frame {FAM131B:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frameshift_status	target_role	breakpoint genomic	breakpoint_exon	breakpoint_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-004484 5-T01-IM6-134	P-00448 45-T01-IM6		GT F2I-BRAF	GT F2I	unknown	threonine - prime	140485631	10	393	True	393	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		GTF2I (NM_032999) - BRAF (NM_004333)fusion: c.768:GTF2I_c.1177+1717:BRAFinv Protein Fusion: mid-exon {GTF2I:BRAF}	NA
EVT-P-000082 8-T05-IM6-135	P-00008 28-T05-IM6		GT F2IRD1-BRAF	GT F2IRD1	in-frame	threonine - prime	140483546	10	393	True	393	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		GTF2IRD1 (NM_005685) - BRAF (NM_004333) rearrangement: c.2320+14147:GTF2IRD1_c.1178-589:BRAFinv Protein Fusion: in frame {GTF2IRD1:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frameshift_status	target_role	breakpoint genomic	breakpoint exon	breakpoint protein position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-004674 5-T01-I M6-136	P-0046745-T01-I M6		KCND2-BRAF	KCND2	out-of-frame	threonine - prime	140485867	10	393	True	393	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KCND2 (NM_012281) - BRAF (NM_004333) fusion: c.1115+91797:KCND2_c.1177+1481:BRAF ^{inv} Protein Fusion: out of frame {KCND2:BRAF}	NA
EVT-P-005695 2-T01-I M6-138	P-0056952-T01-I M6		KIAA1549-BRAF	KIAA1549	in-frame	threonine - prime	140493821	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5248-141:KIAA1549_c.1140+287:BRAF ^{dup} Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0044270-T02-IM6-139	P-0044270-T02-IM6		KIAA1549-BRAF	KIAA1549	in-frame	threonine - prime	140491645	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5247+1560: KIAA1549_c.1140+2463:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0027592-T01-IM6-140	P-0027592-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	threonine - prime	140492343	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused with BRAF exons 9-18) : c.5247+769:KIAA1549_c.1140+1765:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint exon	breakpoint protein position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-005509-T03-IM6-141	P-0055099-T03-IM6		KIAA1549-BRAF	KIAA1549	in-frame	threonine - prime	140490663	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4929+1357: KIAA1549_c.1141-3279:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0058584-T01-IM7-142	P-0058584-T01-IM7		KIAA1549-BRAF	KIAA1549	in-frame	threonine - prime	140491493	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5247+1663: KIAA1549_c.1140+2615:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-006603 0-T01-I M7-143	P-00660 30-T 01-I M7		KIAA1549-BRAF	KIAA1549	in-frame	threonine - prime	140482785	11	439	True	439	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5248-1172: KIAA1549_c.1314+36:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-003515 3-T01-I M6-144	P-00351 53-T 01-I M6		KIAA1549-BRAF	KIAA1549	in-frame	threonine - prime	140491093	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) Fusion (KIAA1549 exon15 fused with BRAF exon9) : c.4930-2046:KIAA1549_c.1140+3015:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint exon	breakpoint protein position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0035153-T03-IM6-145	P-0035153-T03-IM6		KIAA1549-BRAF	KIAA1549	in-frame	threonine - prime	140491093	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4930-2046: KIAA1549_c.1140+3015:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0049925-T01-IM6-146	P-0049925-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	threonine - prime	140489497	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5247+301: KIAA1549_c.1141-2113:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint exon	breakpoint protein position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003584-T03-IM7-147	P-0035844-T03-IM7		KIAA1549-BRAF	KIAA1549	in-frame	threonine - prime	140493285	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4033-1202: KIAA1549_c.1140+823:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-003584-T04-IM7-148	P-0035844-T04-IM7		KIAA1549-BRAF	KIAA1549	in-frame	threonine - prime	140493285	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4033-1202: KIAA1549_c.1140+823:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint exon	breakpoint protein position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-001769 3-T02-I M6-149	P-00176 93-T 02-I M6		KIAA1549-BRAF	KIAA1549	in-frame	threonine - prime	140489537	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) Fusion: c.4930-495: KIAA1549_c.1141-2153:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-006296 3-T01-I M7-150	P-00629 63-T 01-I M7		KIAA1549-BRAF	KIAA1549	in-frame	threonine - prime	140482198	11	439	True	439	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4930-27:KIAA1549_c.1314+623:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0012575-T01-IM5-151	P-0012575-T01-IM5		KIAA1549-BRAF	KIAA1549	in-frame	threonine - prime	140490295	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-15 with BRAF exons 9-18): c.4929+793:KIAA1549_c.1141-2911:BRAFdup Protein fusion: in frame (KIAA1549-BRAF)	NA
EVT-P-0052411-T05-IM7-152	P-0052411-T05-IM7		KIAA1549-BRAF	KIAA1549	in-frame	threonine - prime	140493530	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4930-2296: KIAA1549_c.1140+578:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint exon	breakpoint protein position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0034117-T01-IM6-153	P-0034117-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	threonine - prime	140483812	10	393	True	393	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4346-57:KIAA1549_c.1178-855:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0052411-T03-IM6-154	P-0052411-T03-IM6		KIAA1549-BRAF	KIAA1549	in-frame	threonine - prime	140493530	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4930-2296:KIAA1549_c.1140+578:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0009097-T01-IM5-155	P-0009097-T01-IM5		KIAA1549-BRAF	KIAA1549	in-frame	threonine - prime	140488817	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused in frame with BRAF exons 9-18, which include the BRAF kinase domain); c.5247+3837:KIAA1549_c.1141-1433:BRAFdup Protein fusion: in frame (KIAA1549-BRAF)	NA
EVT-P-0005121-T02-IM5-156	P-0005121-T02-IM5		KIAA1549-BRAF	KIAA1549	in-frame	threonine - prime	140491248	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-15 with BRAF exons 9-18): c.4929+2454:KIAA1549_c.1140+2860:BRADFdup Protein fusion: in frame (KIAA1549-BRAF)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0064872-T01-IM7-157	P-0064872-T01-IM7		KIAA1549-BRAF	KIAA1549	in-frame	threonine - prime	140489277	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4032+4347: KIAA1549_c.1141-1893:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0013777-T01-IM5-158	P-0013777-T01-IM5		KIAA1549-BRAF	KIAA1549	in-frame	threonine - prime	140481674	11	439	True	439	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549(NM_001164665) - BRAF (NM_004333) rearrangement: c.5248-2734:KIAA1549_c.1315-181:BRAFdup Protein fusion: in frame (KIAA1549-BRAF)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0040760-T01-IM6-159	P-0040760-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	threonine - prime	140481540	11	439	True	439	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4929+2627: KIAA1549_c.1315-47:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0014191-T01-IM6-160	P-0014191-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	threonine - prime	140492679	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion : c.5248-320: KIAA1549_c.1140+1429:BRAFdup Protein fusion: in frame (KIAA1549-BRAF)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-001438 1-T01-I M6-161	P-00143 81-T 01-I M6		KIAA1549-BRAF	KIAA1549	in-frame	threonine - prime	140492888	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-15 fused to BRAF exon 9 -18): c.4930-2082:KIAA1549_c.1140+1220:BRAFdup Protein fusion: in frame (KIAA1549-BRAF)	NA
EVT-P-002861 1-T01-I M6-162	P-00286 11-T 01-I M6		KIAA1549-BRAF	KIAA1549	in-frame	threonine - prime	140482786	11	439	True	439	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused to BRAF exons 11-18): c.5248-4149:KIAA1549_c.1314+35:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0015540-T01-IM6-163	P-0015540-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	threonine - prime	140489119	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-15 fused in frame with BRAF exons 9-18, which include the BRAF kinase domain): c.4930-69:KIAA1549_c.1141-1735:BRAFdup Protein fusion: in frame (KIAA1549-BRAF)	NA
EVT-P-0062070-T01-IM7-164	P-0062070-T01-IM7		KIAA1549-BRAF	KIAA1549	in-frame	threonine - prime	140490443	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5248-1064:KIAA1549_c.1141-3059:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint_exon	breakpoint_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0022404-T01-IM6-165	P-0022404-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	threonine - prime	140482177	11	439	True	439	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused with BRAF exons 11-18) : c.5248-2270 :KIAA1549_c.1314+644:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0004032-T01-IM5-166	P-0004032-T01-IM5		KIAA1549-BRAF	KIAA1549	in-frame	threonine - prime	140487892	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) Fusion (KIAA1549 exon 16 fused to BRAF exon 9) : c.5248-1949:KIAA1549_c.1141-508:BRAFdup Protein fusion: in frame (KIAA1549-BRAF)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0043372-T01-IM6-167	P-0043372-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	threonine - prime	140487523	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4929+2204: KIAA1549_c.1141-139:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0028611-T02-IM6-168	P-0028611-T02-IM6		KIAA1549-BRAF	KIAA1549	in-frame	threonine - prime	140482786	11	439	True	439	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused to BRAF exons 11-18): c.5248-4149:KIAA1549_c.1314+35:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0043874-T01-IM6-169	P-0043874-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	threonine - prime	140490001	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RA-S-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5248-2907: KIAA1549_c.1141-2617:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0069024-T04-IM7-170	P-0069024-T04-IM7		KIAA1549-BRAF	KIAA1549	unknown	threonine - prime	140492117	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RA-S-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5059:KIAA1549_c.1140+1991:BRADFdup Protein Fusion: mid-exon {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0029361-T01-IM6-171	P-0029361-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	threonine - prime	140489625	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) Fusion: c.4930-2667; KIAA1549_c.1141-2241:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0016557-T01-IM6-172	P-0016557-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	threonine - prime	140488576	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-15 fused in-frame to BRAF exons 9-18) : c.4929+2955:KIAA1549_c.1141-1192:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003298 2-T01-IM6-173	P-0032982-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	threonine - prime	140491506	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) Fusion (KIAA1549 exons 1-16 fused with BRAF exons 9-18) : c.5248-10:KIAA1549_c.1140+2602:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-006431 0-T04-IM7-174	P-0064310-T04-IM7		KIAA1549-BRAF	KIAA1549	in-frame	threonine - prime	140493446	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5247+3560: KIAA1549_c.1140+662:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0021127-T01-IM6-175	P-0021127-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	threonine - prime	140491741	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-10 fused with BRAF exons 9-18) : c.4032+3996: KIAA1549_c.1140+2367:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0017693-T01-IM6-176	P-0017693-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	threonine - prime	140489537	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (c.1141-2153): c.4930-495:KIAA1549_c.1141-2153:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint_exon	breakpoint_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0020125-T01-IM6-177	P-0020125-T01-IM6		KIAA1549-BRAF	KIAA1549	unknown	threonine - prime	140492175	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RA-S-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-17 fused to BRAF exons 9-18): c.5260:KIAA1549_c.1140+1933:BRAFdup Protein Fusion: mid-exon {KIAA1549:BRAF}	NA
EVT-P-0045683-T02-IM6-178	P-0045683-T02-IM6		KIAA1549-BRAF	KIAA1549	unknown	threonine - prime	140492850	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RA-S-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5189:KIAA1549_c.1140+1258:BRAFdup Protein Fusion: mid-exon {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0019127-T01-IM6-179	P-0019127-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	threonine - prime	140490657	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused to BRAF exons 9-18): c.5247+2110:KIAA1549_c.1141-3273:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0066022-T01-IM7-180	P-0066022-T01-IM7		KIAA1549-BRAF	KIAA1549	in-frame	threonine - prime	140488035	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5248-4074:KIAA1549_c.1141-651:BRAFdup Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0036680-T01-IM6-181	P-0036680-T01-IM6		KLHL12-BRAF	KLHL12	in-frame	threonine - prime	140493984	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KLHL12 (NM_021633) - BRAF (NM_004333) fusion (KLHL12 exons 1-4 fused to BRAF exons 9-18): t(1;7)(q32.1;q34)(chr1:g.202880970::chr7:g.140493984) Protein Fusion: in frame {KLHL12:BRAF}	NA
EVT-P-0047833-T02-IM6-183	P-0047833-T02-IM6		LRBA-BRAF	LRBA	in-frame	threonine - prime	140493444	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		LRBA (NM_001199282) - BRAF (NM_004333) fusion: t(4;7)(q31.3;q34)(chr4:g.151502269::chr7:g.140493444) Protein Fusion: in frame {LRBA:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frameshift_status	target_role	breakpoint genomic	breakpoint exon	breakpoint protein position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-000641 2-T01-I M5-184	P-00064 12-T 01-I M5		LU C7L 2-B RA F	LU C7L 2	unknown	five_prime	140501109	6	287	True	286	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than thisproject's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		LUC7L2 (NM_001244584) - BRAF (NM_004333) Rearrangement : c.1008-2090:LUC7L2_c.860+103:BRAFdel Antisense fusion	NA
EVT-P-002615 8-T04-I M6-185	P-00261 58-T 04-I M6		MAD1L1-BRAF	MAD1L1	in-frame	three_prime	140491749	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MAD1L1 (NM_001013837) - BRAF (NM_004333) fusion: c.1417-9863: MAD1L1_c.1140+2359:BRAFdup Protein Fusion: in frame {MAD1L1:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0026158-T01-IM6-186	P-0026158-T01-IM6		MAD1L1-BRAF	MAD1L1	in-frame	threonine - prime	140491749	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MAD1L1 (NM_001013837) - BRAF (NM_004333) Rearrangement : c.1417-9863_c.1140+2359dup Protein Fusion: in frame {MAD1L1:BRAF}	NA
EVT-P-0006601-T03-IM6-187	P-0006601-T03-IM6		MINDY4-BRAF	MINDY4	in-frame	threonine - prime	140490696	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		FAM188B (NM_032222) - BRAF(NM_004333) Fusion (FAM188B exon5 fused with BRAF exon9) : c.664-366:FAM188B_c.1141-3312:BRAF Protein Fusion: in frame {FAM188B:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint exon	breakpoint protein position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0023220-T01-IM6-188	P-0023220-T01-IM6		MKLN1-BRAF	MKLN1	in-frame	threonine - prime	140496796	8	327	True	327	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MKLN1 (NM_013255) - BRAF (NM_004333) rearrangement: c.98+4070:MKLN1_c.981-2529:BRAFinv Protein Fusion: in frame {MKLN1:BRAF}	NA
EVT-P-0035393-T02-IM6-189	P-0035393-T02-IM6		MKRN1-BRAF	MKRN1	in-frame	threonine - prime	140488436	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MKRN1 (NM_013446) - BRAF (NM_004333) fusion: c.771+833:MKRN1_c.1141-1052: BRAFdup Protein Fusion: in frame {MKRN1:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0009024-T01-IM5-190	P-0009024-T01-IM5		MKRN1-BRAF	MKRN1	in-frame	threonine - prime	140493485	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MKRN1 (NM_013446) - BRAF (NM_004333) rearrangement: c.772-184G>MKRN1_c.1140+623>BRAFdup Protein fusion: in frame (MKRN1-BRAF)	NA
EVT-P-0008083-T03-IM6-191	P-0008083-T03-IM6		MKRN1-BRAF	MKRN1	out-of-frame	threonine - prime	140484511	10	393	True	393	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MKRN1 (NM_013446) - BRAF (NM_004333) rearrangement: c.315-432>MKRN1_c.1178-1554>BRAFdup Protein Fusion: out of frame {MKRN1:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frameshift_status	target_role	breakpoint genomic	breakpoint exon	breakpoint protein position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0001036-T01-IM3-192	P-0001036-T01-IM3		MKRN1-BRAF	MKRN1	in-frame	threonine - prime	140482250	11	439	True	439	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		NA Protein fusion: in frame (MKRN1-BRAF)	NA
EVT-P-0047135-T01-IM6-193	P-0047135-T01-IM6		MKRN1-BRAF	MKRN1	in-frame	threonine - prime	140492510	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MKRN1 (NM_013446) - BRAF (NM_004333) fusion: c.772-972:MKRN1_c.1140+1598: BRAFdup Protein Fusion: in frame {MKRN1:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domains_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0004373-T02-IM5-194	P-0004373-T02-IM5		MKRN1-BRAF	MKRN1	unknown	threonine-primed	140434466	18	744	False	744	True	exact match	lost	0.0	False		Protein kinase domain; RAS-binding domain; Cysteine-rich domain		MKRN1 (NM_013446) - BRAF (NM_004333) rearrangement: c.314+4351:MKRN1_c.232:BRAFdup Protein fusion: mid-exon (MKRN1-BRAF)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0009024-T02-IM6-195	P-0009024-T02-IM6		MKRN1-BRAF	MKRN1	in-frame	threonine - prime	140493485	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MKRN1 (NM_013446) - BRAF (NM_004333) fusion: c.772-185:MKRN1_c.1140+623: BRAFdup Protein Fusion: in frame {MKRN1:BRAF}	NA
EVT-P-0031896-T02-IM6-196	P-0031896-T02-IM6		MKRN1-BRAF	MKRN1	in-frame	threonine - prime	140482581	11	439	True	439	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MKRN1 (NM_013446) - BRAF (NM_004333) fusion (MKRN1 exons 1-4 fused to BRAF exons 11-18): c.772-1018:MKRN1_c.1314+240: BRAFdup Protein Fusion: in frame {MKRN1:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frameshift_status	target_role	breakpoint genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-000808 3-T02-IM6-197	P-0008083-T02-IM6		MKRN1-BRAF	MKRN1	out-of-frame	threonine - prime	140484505	10	393	True	393	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MKRN1 (NM_013446) - BRAF (NM_004333) fusion (MKRN1 exons 1-2 fused with BRAF exons 10-18): c.315-428:MKRN1_c.1178-1548:BRAFdup Protein Fusion: out of frame {MKRN1:BRAF}	NA
EVT-P-001353 6-T01-IM5-198	P-0013536-T01-IM5		MKRN1-BRAF	MKRN1	in-frame	threonine - prime	140483618	10	393	True	393	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MKRN1 (NM_013446) - BRAF (NM_004333) rearrangement: c.544+172:MKRN1_c.1178-661:BRAFdup Protein fusion: in frame (MKRN1-BRAF)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint exon	breakpoint protein position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-004803 6-T01-IM6-199	P-0048036-T01-IM6		NEO1-BRAF	NEO1	in-frame	threonine - prime	140497651	8	327	True	327	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		NEO1 (NM_002499) - BRAF (NM_004333)fusion: t(7;15)(q34;q24.1)(chr7:g.140497651::chr15:g.73575682) Protein Fusion: in frame {NEO1:BRAF}	NA
EVT-P-004397 2-T03-IM6-200	P-0043972-T03-IM6		NRF1-BRAF	NRF1	in-frame	threonine - prime	140487712	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		NRF1 (NM_005011) - BRAF (NM_004333)fusion: c.607-7410:NRF1_c.1141-328:BRAFinv Protein Fusion: in frame {NRF1:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint exon	breakpoint position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-006544 6-T01-I M7-201	P-00654 46-T 01-I M7		NRF1-BRAF	NRF1	in-frame	threonine - prime	140483810	10	393	True	393	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		NRF1 (NM_005011) - BRAF (NM_004333)fusion: c.1349-10893:NRF1_c.1178-853:BRAFin v Protein Fusion: in frame {NRF1:BRAF}	NA
EVT-P-005900 2-T01-I M7-202	P-00590 02-T 01-I M7		OSBPL7-BRAF	OSBPL7	in-frame	threonine - prime	140491530	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		OSBPL7 (NM_145798) - BRAF (NM_004333) fusion: t(7;17)(q34;q21.32)(chr7:g.140491530::chr17:g.45890381) Protein Fusion: in frame {OSBPL7:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0054958-T02-IM6-203	P-0054958-T02-IM6		PAK2-BRAF	PAK2	in-frame	threonine - prime	140486065	10	393	True	393	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		PAK2 (NM_002577) - BRAF (NM_004333) rearrangement: t(3;7)(q29;q34)(chr3:g.196530579::chr7:g.140486065) Protein Fusion: in frame {PAK2:BRAF}	NA
EVT-P-0008789-T01-IM5-205	P-0008789-T01-IM5		PHTF2-BRAF	PHTF2	in-frame	threonine - prime	140489236	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		PHTF2 (NM_001127358) - BRAF (NM_004333) rearrangement: c.224-367:PHTF2_c.1141-1852:BRAFInv Protein fusion: in frame (PHTF2-BRAF)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frameshift_status	target_role	breakpoint genomic	breakpoint exon	breakpoint protein position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0044819-T01-IM6-206	P-0044819-T01-IM6		PLP1-BRAF	PLP1	out-of-frame	threonine - prime	140482123	11	439	True	439	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		PPAP2A (NM_176895) - BRAF (NM_004333) fusion: t(5;7)(q11.2;q34)(chr5:g.54789192::chr7:g.140482123) Protein Fusion: out of frame {PPAP2A:BRAF}	NA
EVT-P-0009694-T01-IM5-207	P-0009694-T01-IM5		PRKAR1B-BRAF	PRKAR1B	in-frame	threonine - prime	140491855	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		PRKAR1B (NM_001164758) -BRAF (NM_004333) Rearrangement : c.892-2551:PRKAR1B_c.1140+2253:BRAFdup Protein fusion: in frame {PRKAR1B-BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint exon	breakpoint protein position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0008567-T01-IM5-208	P-0008567-T01-IM5		PRKAR2B-BRAF	PRKAR2B	in-frame	threonine - prime	140486416	10	393	True	393	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		PRKAR2B (NM_002736) - BRAF (NM_004333) rearrangement: c.307+8063:PRKAR2B_c.1177+932:BRAFinv Protein fusion: in frame (PRKAR2B-BRAF)	NA
EVT-P-0041181-T01-IM6-209	P-0041181-T01-IM6		PTPRZ1-BRAF	PTPRZ1	in-frame	threonine - prime	140492428	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		PTPRZ1 (NM_002851) - BRAF (NM_004333) fusion: c.5367+457:PTPRZ1_c.1140+1680:BRAFinv Protein Fusion: in frame {PTPRZ1:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint exon	breakpoint protein position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0012884-T01-IM5-210	P-0012884-T01-IM5		PWWP2A-BRAF	PWWP2A	in-frame	threonine - prime	140495243	8	327	True	327	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		PWWP2A (NM_001130864) - BRAF(NM_004333) Rearrangement : t(5;7)(q33.3;q34)(chr5:g.159539324::chr7:g.140495243) Protein fusion: in frame (PWWP2A-BRAF)	NA
EVT-P-0068260-T01-IM7-211	P-0068260-T01-IM7		RBMS3-BRAF	RBMS3	in-frame	threonine - prime	140482707	11	439	True	439	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		RBMS3 (NM_001003793) - BRAF (NM_004333) fusion: t(3;7)(p24.1;q34)(chr3:g.30030473::chr7:g.140482707) Protein Fusion: in frame {RBMS3:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint exon	breakpoint protein position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0064764-T01-IM7-212	P-0064764-T01-IM7		RGS3-BRAF	RGS3	out-of-frame	threesome_prime	140493282	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		RGS3 (NM_144488) - BRAF (NM_004333)fusion: t(7;9)(q34;q32)(chr7:g.140493282::chr9:g.116291333) Protein Fusion: out of frame {RGS3:BRAF}	NA
EVT-P-0041308-T02-IM6-213	P-0041308-T02-IM6		RRBP1-BRAF	RRBP1	in-frame	threesome_prime	140485917	10	393	True	393	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		RRBP1 (NM_004587) - BRAF (NM_004333) fusion: t(7;20)(q34;p12.1)(chr7:g.140485917::chr20:g.17634546) Protein Fusion: in frame {RRBP1:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-000739 2-T01-I M5-214	P-00073 92-T 01-I M5		SCRIB-BRAF	SCRIB	in-frame	threonine - prime	140482150	11	439	True	439	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SCRIB (NM_182706) - BRAF (NM_004333) rearrangement: t(7;8)(q34;q24.3)(chr7:g.140482150::chr8:g.144876675) Protein fusion: in frame (SCRIB-BRAF)	NA
EVT-P-005663 9-T01-I M6-215	P-00566 39-T 01-I M6		SETBP1-BRAF	SETBP1	in-frame	threonine - prime	140486594	10	393	True	393	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SETBP1 (NM_015559) - BRAF (NM_004333) fusion: t(7;18)(q34;q12.3)(chr7:g.140486594::chr18:g.42638299) Protein Fusion: in frame {SETBP1:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frameshift_status	target_role	breakpoint genomic	breakpoint exon	breakpoint protein position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0048950-T01-IM6-216	P-0048950-T01-IM6		SN-D1-BRAF	SN-D1	in-frame	threonine - prime	140492033	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333)fusion: c.1152+30037:SND1_c.1140+2075:BRAFinv Protein Fusion: in frame {SND1:BRAF}	NA
EVT-P-0012395-T01-IM5-217	P-0012395-T01-IM5		SN-D1-BRAF	SN-D1	in-frame	threonine - prime	140493527	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) rearrangement: c.1152+27174:SND1_c.1140+581:BRAFinv Protein fusion: in frame (SND1-BRAF)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0009619-T01-IM5-218	P-0009619-T01-IM5		SN-D1-BRAF	SN-D1	in-frame	threonine - prime	140482494	11	439	True	439	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) rearrangement: c.1968+2394:SND1_c.1314+327:BRAFinv Protein fusion: in frame (SND1-BRAF)	NA
EVT-P-0052299-T04-IM6-219	P-0052299-T04-IM6		SN-D1-BRAF	SN-D1	unknown	threonine - prime	140482935	10	400	False	400	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion: c.1038+2929:SND1_c.1200:BRAFinv Protein Fusion: mid-exon {SND1:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint exon	breakpoint protein position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0001687-T01-IM3-220	P-0001687-T01-IM3		SN-D1-BRAF	SN-D1	in-frame	threonine - prime	140490224	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) inversion: c.1152+28381_c.1141-2840inv Protein fusion: in frame (SND1-BRAF)	NA
EVT-P-0004103-T01-IM5-221	P-0004103-T01-IM5		SN-D1-BRAF	SN-D1	in-frame	threonine - prime	140481685	11	439	True	439	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) rearrangement: c.1038+5799:SND1_c.1315-192:BRAFinv. Protein fusion: in frame (SND1-BRAF)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint exon	breakpoint protein position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0029727-T01-IM6-222	P-0029727-T01-IM6		SN-D1-BRAF	SND1	in-frame	threonine - prime	140482257	11	439	True	439	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333)fusion: c.1152+12095:SND1_c.1314+564:BRAF in frame {SND1:BRAF}	NA
EVT-P-0068194-T01-IM7-223	P-0068194-T01-IM7		SN-D1-BRAF	SND1	in-frame	threonine - prime	140489175	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333)fusion: c.1152+17686:SND1_c.1141-1791:BRAF in frame {SND1:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frameshift_status	target_role	breakpoint genomic	breakpoint_exon	breakpoint_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0008140-T01-IM5-224	P-0008140-T01-IM5		SN-D1-BRAF	SND1	in-frame	threonine - prime	140489439	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion (SND1 exons 1-9 fused with BRAF exons 9-18) : c.1039-3849:SND1_c.1141-2055:BRAFinv Protein fusion: in frame (SND1-BRAF)	NA
EVT-P-0034168-T01-IM6-225	P-0034168-T01-IM6		SN-D1-BRAF	SND1	in-frame	threonine - prime	140489421	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion (SND1 exons 1-10 fused to BRAF exons 9-18): c.1152+6972:SND1_c.1141-2037:BRAFinv Protein Fusion: in frame {SND1:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint_exon	breakpoint_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0025075-T01-IM6-226	P-0025075-T01-IM6		SN-D1-BRAF	SND1	in-frame	threonine - prime	140490546	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion (SND1 exons 1- 9 with BRAF exons 9-18) : c.1039-2471: SND1_c.1141-3162: BRAFInV Protein Fusion: in frame {SND1:BRAF}	NA
EVT-P-0002071-T02-IM5-227	P-0002071-T02-IM5		SN-D1-BRAF	SND1	in-frame	threonine - prime	140489292	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion (SND1 exons 1-10 fused with BRAF exons 9-18) : c.1152 +18606:SND1_c.1141-1908:BRAFInV Protein fusion: in frame (SND1-BRAF)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0002197-T01-IM3-228	P-0002197-T01-IM3		SN-D1-BRAF	SN-D1	in-frame	five_prime	140488303	8	380	True	380	True	exact match	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		c.1528-8436:SN-D1_c.1141-922:BRAFInv Protein fusion: in frame (BRAF-SND1)	NA
EVT-P-0007533-T01-IM5-229	P-0007533-T01-IM5		SN-D1-BRAF	SN-D1	in-frame	three_prime	140482226	11	439	True	439	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion (SND1 exons 1-10 with BRAF exons 11-18) : c.1152+31738:SND1_c.1314+595:BRAFInv Protein fusion: in frame (SND1-BRAF)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint_exon	breakpoint_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0036567-T01-IM6-230	P-0036567-T01-IM6		SN-D1-BRAF	SN-D1	in-frame	threonine - prime	140493552	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) Fusion (SND1 exon10 fused with BRAF exon9) ; c.1153-22013:SND1_c.1140+556:BRAFInv Protein Fusion: in frame {SND1:BRAF}	NA
EVT-P-0034845-T01-IM6-231	P-0034845-T01-IM6		SN-D1-BRAF	SN-D1	in-frame	threonine - prime	140489241	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) rearrangement: c.1527+3658:SND1_c.1141-1857:BRAFInv Protein Fusion: in frame {SND1:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint exon	breakpoint protein position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-005436 4-T01-I M6-232	P-00543 64-T 01-I M6		SNX8-BRAF	SNX8	in-frame	threonine - prime	140488730	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SNX8 (NM_013321) - BRAF (NM_004333)fusion: c.1284+565:SNX8_c.1141-1346:BRAFdup Protein Fusion: in frame {SNX8:BRAF}	NA
EVT-P-002677 3-T01-I M6-234	P-00267 73-T 01-I M6		SVOPL-BRAF	SVOPL	unknown	five - prime	140482097	10	438	True	438	True	exact match	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) rearrangement: c.1315-604:BRAF_chr7:g.138277426del Transcript Fusion {BRAF:SVOPL}	NA
EVT-P-005121 4-T01-I M6-235	P-00512 14-T 01-I M6		TMPRSS2-BRAF	TMPRSS2	in-frame	threonine - prime	140502592	6	238	True	238	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain	Cysteine-rich domain	TMPRSS2 (NM_001135099) - BRAF (NM_004333) fusion: t(7;21)(q34;q22.3)(chr7:g.140502592::chr21:g.42867846) Protein Fusion: in frame {TMPRSS2:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0018524-T01-IM6-236	P-0018524-T01-IM6		TM PRSS2-BRAF	TM PRSS2	out-of-frame	threonine - prime	140490337	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TMPRSS2 (NM_001135099) - BRAF (NM_004333) Rearrangement : t(21;7)(q22.3;q34)(chr21:g.42873801::chr7:g.140490337) Protein Fusion: out of frame {TMPRSS2:BRAF}	NA
EVT-P-0022778-T01-IM6-237	P-0022778-T01-IM6		TM PRSS2-BRAF	TM PRSS2	in-frame	threonine - prime	140483147	10	393	True	393	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TMPRSS2 (NM_001135099) - BRAF (NM_004333) Rearrangement : t(7;21)(q34;q22.3)(chr7:g.140483147::chr21:g.42878796) Protein Fusion: in frame {TMPRSS2:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint exon	breakpoint protein position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-004383 6-T01-IM6-239	P-0043836-T01-IM6		TRA2B-BRAF	TRA2B	in-frame	threonine - prime	140491978	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TRA2B (NM_004593) - BRAF (NM_004333) fusion: t(3;7)(q27.2;q34)(chr3:g.185650761::chr7:g.140491978) Protein Fusion: in frame {TRA2B:BRAF}	NA
EVT-P-005443 0-T02-IM6-240	P-0054430-T02-IM6		TRIM24-BRAF	TRIM24	in-frame	threonine - prime	140487875	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TRIM24 (NM_015905) - BRAF (NM_004333) fusion: c.2719-489:T TRIM24_c.1141-491: BRAFInv Protein Fusion: in frame {TRIM24:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint exon	breakpoint protein position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-002569 9-T02-IM6-241	P-0025699-T02-IM6		TRIM24-BRAF	TRIM24	in-frame	threonine - prime	140486854	10	393	True	393	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TRIM24 (NM_015905) - BRAF (NM_004333) fusion: c.2014+674:T RIM24_c.1177+494: BRAFInv Protein Fusion: in frame {TRIM24:BRAF}	NA
EVT-P-004511 1-T01-IM6-242	P-0045111-T01-IM6		TRIM24-BRAF	TRIM24	in-frame	threonine - prime	140488049	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TRIM24 (NM_015905) - BRAF (NM_004333) fusion: c.2257-760:T RIM24_c.1141-665: BRAFInv Protein Fusion: in frame {TRIM24:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint exon	breakpoint protein position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-006683 5-T01-I M7-243	P-00668 35-T 01-I M7		TRIM24-BRAF	TRIM24	in-frame	threonine - prime	140490589	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TRIM24 (NM_015905) - BRAF (NM_004333) fusion: c.1879-502:T RIM24_c.1141-3205: BRAFInv Protein Fusion: in frame {TRIM24:BRAF}	NA
EVT-P-006845 1-T01-I M7-244	P-00684 51-T 01-I M7		TRIM24-BRAF	TRIM24	in-frame	threonine - prime	140482765	11	439	True	439	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TRIM24 (NM_015905) - BRAF (NM_004333) fusion: c.484-860:TRIM24_c.1314+56:BR AFInv Protein Fusion: in frame {TRIM24:BRAF}	NA

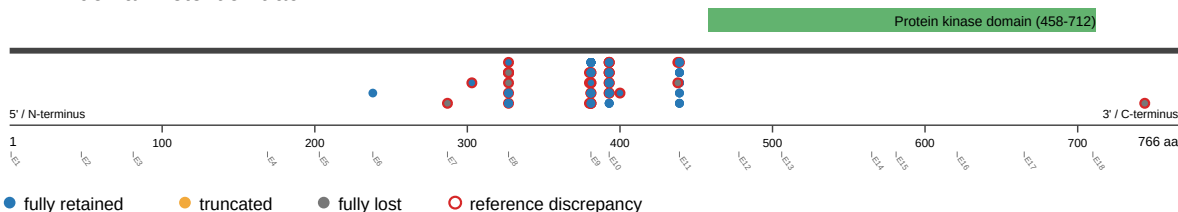
event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003098-T01-IM6-245	P-0030988-T01-IM6		TRIM24-BRAF	TRIM24	unknown	threonine - prime	140491002	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TRIM24 (NM_015905) - BRAF (NM_004333) fusion: c.2349:TRIM24_c.1140+3106:BRAFinv Protein Fusion: mid-exon {TRIM24:BRAF}	NA
EVT-P-0032569-T01-IM6-246	P-0032569-T01-IM6		TRIM24-BRAF	TRIM24	in-frame	threonine - prime	140487936	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TRIM24 (NM_015905) - BRAF (NM_004333) Rearrangement : c.1878+791:TRIM24_c.1141-552inv Protein Fusion: in frame {TRIM24:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint genomic	breakpoint exon	breakpoint protein position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003256 9-T02-IM6-247	P-0032569-T02-IM6		TRIM24-BRAF	TRIM24	in-frame	threonine - prime	140487936	9	381	True	381	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TRIM24 (NM_015905) - BRAF (NM_004333) fusion: c.1878+791:TRIM24_c.1141-552: BRAFInv Protein Fusion: in frame {TRIM24:BRAF}	NA
EVT-P-001903 9-T01-IM6-248	P-0019039-T01-IM6		UBE2H-BRAF	UBE2H	out-of-frame	threonine - prime	140499248	8	327	True	327	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		BRAF (NM_004333) - UBE2H (NM_003344) rearrangement: c.980+914:BRAF_c.299-4611:UBE2Hdup Protein Fusion: out of frame {UBE2H:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	fragment_status	target_role	breakpoint genomic	breakpoint exon	breakpoint protein position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-002776 2-T01-IM6-250	P-0027762-T01-IM6		ZC3HAV1-BRAF	ZC3HAV1	in-frame	threonine - prime	140485946	10	393	True	393	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		ZC3HAV1 (NM_020119) - BRAF (NM_004333) rearrangement: c.698-431:ZC3HAV1_c.1177+1402:BRAFdup Protein Fusion: in frame {ZC3HAV1:BRAF}	NA
EVT-P-000892 0-T01-IM5-251	P-0008920-T01-IM5		ZNF207-BRAF	ZNF207	in-frame	threonine - prime	140485847	10	393	True	393	True	exact match	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		ZNF207 (NM_001098507) - BRAF (NM_004333) rearrangement: t(7;17)(q34;q11.2)(chr7:g.140485847::chr17:g.30687533) Protein fusion: in frame (ZNF207-BRAF)	NA

domain retention outliers

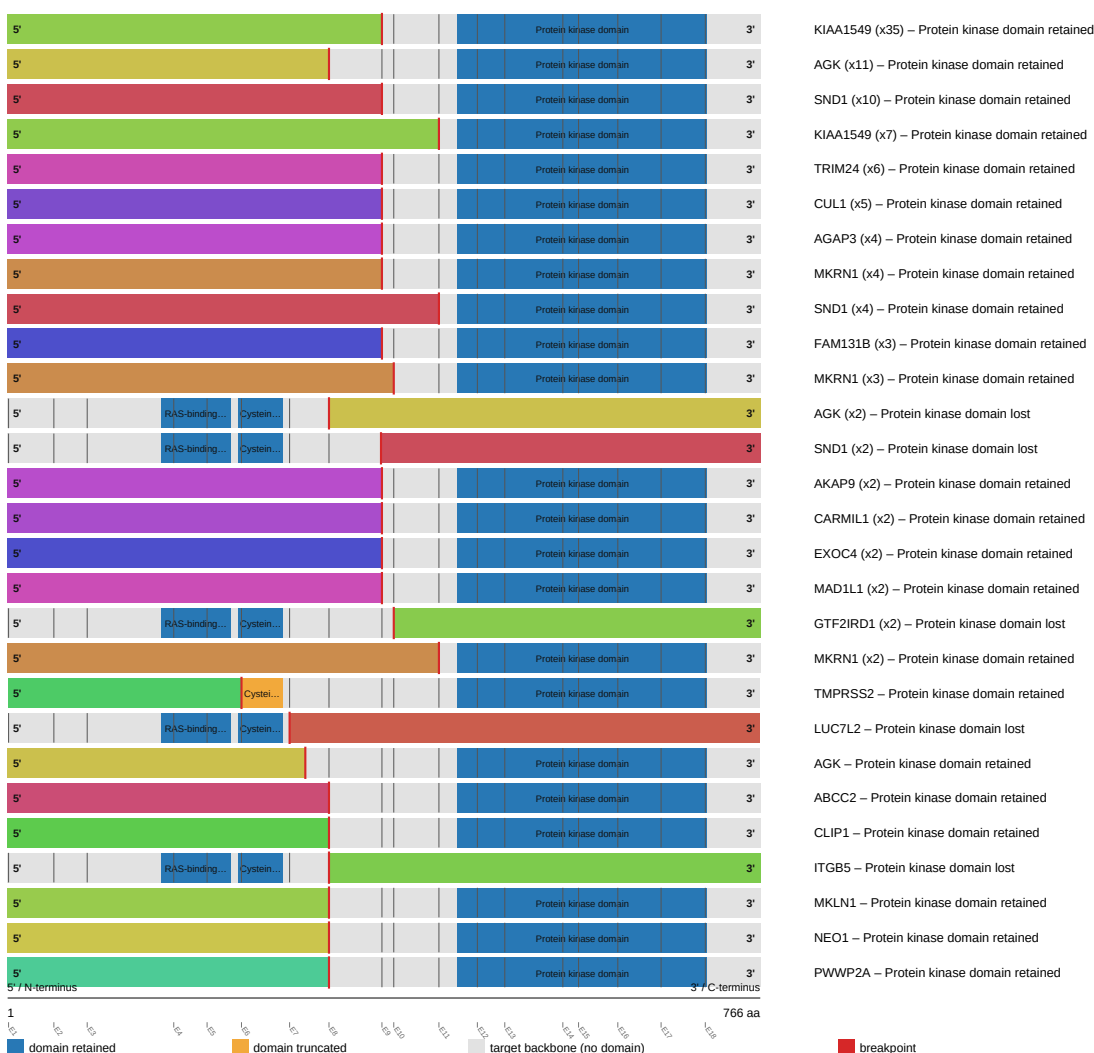
BRAF domain-retention track



fusion schematic

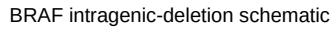
BRAF fusion-transcript schematic

Fusion order varies by row; follow each row's 5' and 3' labels (BRAF portion is domain-colored, exon ticks)

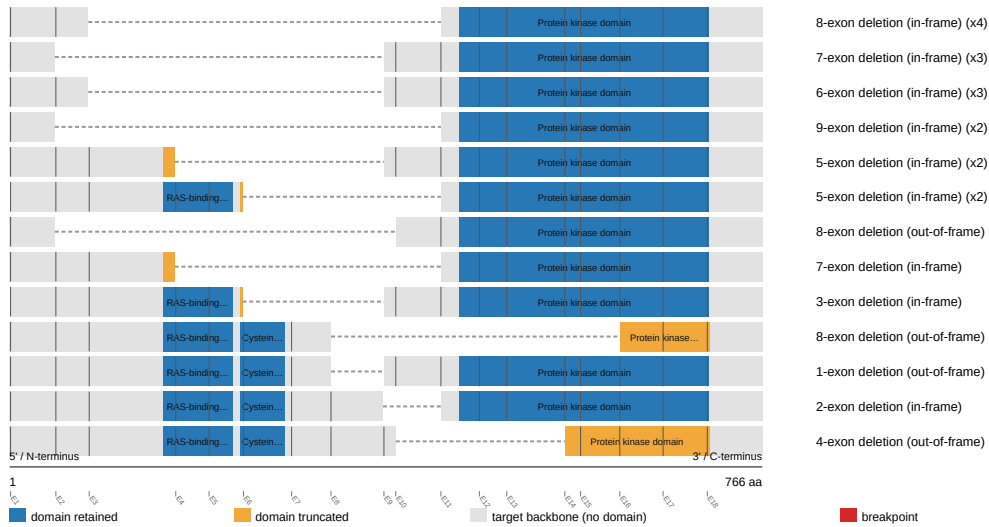


Showing the top 28 of 89 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

intragenic deletion schematic



retained N-terminal block → deleted span (plain connector) → resumed C-terminal block



reference comparison

Reference vs msk_impact_50k_2026

